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MODULE: CM3015 Machine Learning and Neural Networks

Sentiment Classification Using Dense Neural Networks (IMDB Dataset)

Abstract

This project applies the universal deep learning workflow [1] to a supervised binary sentiment classification task using the IMDB movie review dataset [3]. The objective is to develop, evaluate, and interpret fully connected neural network models under strict architectural constraints: TensorFlow Sequential models using only Dense and Dropout layers. Reviews are vectorized into fixed-length multi-hot (bag-of-words) representations based on the most frequent tokens, enabling the use of feedforward networks while maintaining a transparent link between input preprocessing choices and model behavior.

A disciplined evaluation protocol is followed throughout: the training data is partitioned into training and validation subsets for model selection and hyperparameter decisions, while the held-out test set is reserved exclusively for final performance reporting. The investigation proceeds systematically from a simple baseline model to a higher-capacity model to intentionally increase model complexity and observe overfitting behavior. This is followed by regularization and hyperparameter tuning, including dropout rates of 0.5 and 0.3, with a learning rate of 0.0005, to improve generalization. Results are interpreted using learning curves (training vs validation loss/accuracy) and comparative summaries, with the focus on explaining performance changes in terms of the underfitting overfitting trade off rather than only reporting final scores.

The final outcome provides evidence based conclusions about model capacity, overfitting dynamics, and the effectiveness of regularization strategies for text sentiment classification using foundational deep learning tools.

1. Introduction & Problem Definition

This project looks it the task of binary sentiment classification of movie reviews, where the dataset objective is to predict whether a given review expresses a positive or negative sentiment, based on its text content. Each review is an independent sample, and the model is trained on labeled examples and evaluated on previously unseen data.

Input (X): Textual movie reviews

Target (y): Binary sentiment label (positive or negative)

Learning Type: Supervised learning

Problem Type: Binary classification

In alignment with the universal deep learning workflow in DLWP Section 4.5[1], this project functions under two key hypotheses:

Sentiment is predictable from textual features: The review's text contains sufficient information to infer whether the sentiment is positive or negative.

The labeled data is sufficiently informative: The available labeled examples are adequate for a neural network to learn meaningful patterns that generalize to unseen data.

These hypotheses will be assessed in an indirect manner. In particular, we will evaluate whether the model's validation performance significantly exceeds the baseline and investigate its generalization by comparing the performance gaps between training and validation data. A well generalizing model will exhibit a small gap, suggesting strong performance on both the training set and unseen data.

The primary objective of this project is to explore the relationship between model capacity and regularization techniques in affecting model performance and generalization. Following the approach in DLWP 4.5, the investigation proceeds as follows:

Establish a baseline model to set a starting performance benchmark.

Increase model capacity intentionally to observe overfitting, testing the limits of the model's ability to memorize training data.

Apply regularization techniques, such as dropout with rates of 0.5 and 0.3, and hyperparameter tuning (learning rate of 0.0005) to reduce overfitting, while maintaining strong performance on validation data.

By changing these factors, we can better understand the dynamics between model complexity and its ability to generalize to new and unseen data.

2. Dataset selection and justification

The dataset used in this project is the IMDB movie review dataset, which is available through TensorFlow/Keras [3]. It contains 50000 labeled movie reviews, with the equal distribution of positive and negative sentiment classes. The dataset is split into 25000 training samples and 25000 test samples, making it a balanced and sufficiently large dataset for supervised binary classification.

This dataset is suitable for the project for several reasons. Firstly, its balanced class distribution ensures that performance can be evaluated using standard classification metrics like accuracy without issues related to class imbalance, making comparisons between model configurations more understandable. Secondly, the dataset is large enough to support systematic experimentation with different model capacities, allowing us to observe situations like underfitting, overfitting, and the effects of regularization, in accordance with the workflow outlined in DLWP Section 4.5 [1].

Finally, the IMDB reviews can be vectorized into fixed length multi hot (bag of words) representations, which are directly compatible with the fully connected Dense networks specified in the project's architectural restrictions. This eliminates the need for models such as CNNs or RNNs. The dataset is standardized and easily accessible via Keras, enabling the focus of the project to remain on controlled experimentation with model design, training behavior, and generalization performance, rather than data collection or complex preprocessing pipelines.

3. Measure of Success and Evaluation Protocol

3.1 Measure of Success

Measure of success is important in machine learning because it determines how model performance is compared across experiments and informs modeling decisions [1]. In this project, the task is to classify IMDB movie reviews as either positive or negative, and the dataset is balanced across the two classes. Due to this reason, accuracy is used as the main evaluation metric, as it provides an interpretable measure of a set of correctly classified reviews and enables straightforward comparison between different model configurations.

Accuracy Formula:

Accuracy is computed as the ratio of correctly predicted instances (both positive and negative reviews) to the total number of instances:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

[7][5]

Where:

TP = True Positives (positive reviews correctly predicted as positive)

TN = True Negatives (negative reviews correctly predicted as negative)

FP = False Positives (negative reviews incorrectly predicted as positive)

FN = False Negatives (positive reviews incorrectly predicted as negative)

Given the balanced class distribution, performance meaningfully above a simple baseline (approximately 50% accuracy) indicates that the model is learning useful patterns.

However, the primary aim of this study is not solely to maximize accuracy but to investigate how model capacity and regularization affect generalization. Therefore, success is defined using two complementary criteria:

A model should achieve strong validation accuracy, indicating effective performance on unseen data during development.

Model behavior should demonstrate improved generalization as regularization is introduced, which is assessed by monitoring the generalization gap between training and validation performance.

In practice, this involves tracking both training and validation curves. A growing gap between training and validation accuracy as model capacity increases is treated as evidence of overfitting, while a reduced gap (without substantial loss of validation performance) is treated as improved generalization.

Binary Cross-Entropy Loss Formula:

In addition to accuracy, training and validation loss values are monitored throughout all experiments. For binary classification, the model is trained by optimizing binary cross-entropy loss, which quantifies the error between the predicted probabilities and the true binary labels (positive or negative).

The binary cross-entropy loss function is given by:

$$\mathrm{BCE} = -\frac{1}{N}\sum_{i=1}^{N}\left[y_i \log(p_i) + (1-y_i)\log(1-p_i)\right]$$

[6][8] Where:

yi : is the true binary label (0 or 1) for the i-th sample,

pi : is the predicted probability of the positive class for the i-th sample,

N : is the total number of samples.

Changes in loss often provide earlier indications of overfitting than accuracy alone. During model development, validation metrics (accuracy and loss) are used for model selection and hyperparameter decisions, while the held-out test set is reserved for final evaluation only. Taken together, validation accuracy, loss curves, and the generalization gap provide the proper definition of success. This supports the projects objective of systematically analyzing

the trade off between fitting the training data and generalizing to unseen data when varying model complexity and applying regularization.

3.2 Evaluation protocol

To measure progress reliably and study generalization as model capacity and regularization change, this project adopts a hold-out validation protocol, consistent with the approach recommended when sufficient data is available. The IMDB dataset, provided through Keras, includes predefined training and test sets. The test set is kept aside for final evaluation and is not used during model development, architectural decisions, or hyperparameter tuning.

During development, the original training set is further divided into two parts:

Training subset: Used to fit model parameters (weights) using gradient based optimization.

Validation subset: Used to evaluate performance during training and to guide decisions about model capacity and regularization.

This separation is important because the goal of the experimentation is to observe how models behave as they move from underfitting toward overfitting and then toward improved generalization through regularization. Validation performance provides the feedback required to detect overfitting when training performance continues to improve while validation performance stagnates or degrades and to compare model configurations fairly.

To ensure meaningful comparisons across experiments, the validation split is kept the same across all runs (the same training validation partition is used when varying the number of units,layers, dropout rates, or training duration. This prevents results from being affected by variability introduced through different validation samples. Reproducibility is further ensured by fixing random seeds where applicable and by reporting key training settings such as batch size and number of epochs for each experiment.

Performance is evaluated once on the held out test set, only after a final model configuration has been selected based on validation results. This final test evaluation provides an unbiased estimate of real-world generalization performance and helps identify whether repeated tuning has led to overfitting to the validation process.

4. Data Preparation

Before a neural network can be trained, the raw input must be converted into numerical tensors that are compatible with the chosen model class and with gradient based optimization. In this project, the IMDB reviews are naturally textual, but the modeling constraint is to use only fully connected networks built from Dense and Dropout layers [1]. Fully connected networks require each sample to have the same fixed dimensionality, so the

text data must be transformed into a fixed length numerical representation that can be used directly as input features.

The IMDB dataset is loaded using the TensorFlow/Keras API [4]. Each review is encoded as a variable length sequence of integers, where each integer corresponds to a word index in a predefined vocabulary. To control input dimensionality and the number of trainable parameters in the first layer, the vocabulary is restricted to the 10000 most frequent word indices. This choice reduces computational cost, limits sparsity, and supports clearer analysis of overfitting by reducing the reliance on very rare words(tokens).

A consequence of this choice is that some rare words, which may carry sentiment information, are discarded or mapped outside the retained vocabulary. This limitation is acceptable in the present study because the objective is not to maximize performance through elaborate feature engineering, but to examine optimization and generalization behavior under controlled changes in model capacity and regularization.

4.1 Loading the Dataset

A hold-out validation set is created from the original training data to assess generalization during model development and to support systematic comparisons across experiments. To ensure that performance differences can be attributed to changes in architecture, capacity, or regularization rather than to changes in validation samples, the same validation partition is retained across all runs. Prior to the split, the training data is shuffled once using a fixed random seed to reduce any ordering bias and to support reproducibility.

In [12]:
```
# import the tensorflow library used to build and train machine learning models
import tensorflow as tf

# import the imdb dataset loader from keras which has movie reviews used for sentim
from tensorflow.keras.datasets import imdb

# import numpy for numerical operations and handling arrays
import numpy as np
```

In [86]:
```
# limits the vocabulary to the top 10000 most common words[1]
num_words = 10000

# load the imdb dataset
# x_train and y_train contain the training reviews and their labels
# x_test and y_test contain the testing reviews and their labels
# the num_words makes sure that only the top 10000 most frequent words are included
(x_train, y_train), (x_test, y_test) = imdb.load_data(num_words=num_words)
```

In [87]:
```
# print the test and train samples and lables
print("Training samples:", len(x_train))
print("Training labels:", len(y_train))
```

```
print("Test samples:", len(x_test))
print("Test labels:", len(y_test))
```

```
Training samples: 25000
Training labels: 25000
Test samples: 25000
Test labels: 25000
```

At this stage, the inputs are still sequences of word indices with different lengths. While such sequences are appropriate for models that handle temporary structure, they cannot be fed directly into the Dense layers because the input dimension must be fixed. The next step is to convert each review into a fixed length vector representation.

4.2 Converting Reviews to Fixed-Length Vectors

To make the reviews compatible with fully connected networks, each review is converted into a fixed-length binary multi-hot (bag-of-words) representation. The vocabulary is restricted to the 10000 most frequent word indices in order to control input dimensionality and the number of trainable parameters in the first layer. This choice reduces computational cost, limits sparsity, and makes the effect of model capacity easier to interpret across experiments.

Each review is represented as a binary vector of length 10000, where each position indicates whether the corresponding word appears anywhere in the review. A binary multi-hot representation is used rather than raw word counts in order to keep feature values bounded and reduce sensitivity to repeated word frequency. This makes the representation less affected by review length than count-based representations. Although this representation ignores word order and local context, it is appropriate for this project because it remains within the Dense/Dropout network restriction while preserving a meaningful signal for sentiment classification.

Example of Multi-Hot Vectorization:

For example, the review "The movie was great" might be converted into the following binary vector:

[0,1,1,0,0,0,...,1,0,0]

This vector represents the presence or absence of specific words in the vocabulary.

In [90]:
```
# define a function to convert sequences of word indices into a binary vector repre
def vectorize_sequences(sequences, dimension):

    # create a matrix filled with zeros
    # the number of rows equals the number of sequences and the number of columns e
    results = np.zeros((len(sequences), dimension), dtype=np.float32)

    # loop through each sequence and its index
    for i, sequence in enumerate(sequences):

        # set the positions of the words that appear in the sequence to 1.0
```

```
        # this creates a multi-hot encoded vector representation of the review
        results[i, sequence] = 1.0

    # return the vectorized result
    return results


# convert the training sequences into vector form using the vocabulary size
x_train_vec = vectorize_sequences(x_train, num_words)

# convert the test sequences into vector form using the same vocabulary size
x_test_vec = vectorize_sequences(x_test, num_words)

# print the shape of the vectorized training data to verify its dimensions
print(x_train_vec.shape)

# print the shape of the vectorized test data to verify its dimensions
print(x_test_vec.shape)
```

```
(25000, 10000)
(25000, 10000)
```

After vectorization, the input data becomes a two dimensional tensor. In the original Keras split, x_train_vec has shape of 25000, 10000 and x_test_vec has shape of 25000, 10000. Every row represents one review, and each column corresponds to a word in the restricted vocabulary. Like this, every review is converted into a fixed length numerical vector that can be passed into Dense layers.

This approach ignores word order, sentence structure, and contextual relationships. While this means the model cannot capture more complex language patterns, that limitation is acceptable for this project. The goal is to analyze how changes in model capacity and regularization affect optimization and generalization within a simple, controlled architecture. Because the representation is fixed-length, it supports fair and repeatable comparisons across model configurations without introducing more advanced sequence models that are beyond the scope of the module.

4.3 Preparing the Labels

The sentiment labels are binary values, where 0 means negative review and 1 means positive review. These labels are converted to float32 format to ensure compatibility with gradient based optimization and the binary cross entropy loss function used during training. Binary cross entropy loss operates on the predicted probabilities (floating point values) and the true binary labels, making it important for the labels to also be in floating point format.

In [92]:
```
# make the y train and test float
y_train = np.asarray(y_train).astype("float32")
y_test = np.asarray(y_test).astype("float32")
```

After conversion, the labels remain one-dimensional target arrays, with one binary value for each review.For the shape of both y_train and y_test, each entry corresponds to the

sentiment label for the respective review. This representation is appropriate for supervised binary classification and matches the output structure used by the models in later sections.

If the labels were initially in text format (positive,negative), they would be encoded into numeric binary labels (0,1) during preprocessing.

For multi class classification tasks, labels are typically one hot encoded, where each class is represented by a vector of binary values.

4.4 Creating a Validation Set

A hold out validation set is created from the original training data to assess generalization during model development and to support systematic comparisons across experiments. Hold out validation was chosen because it is efficient and appropriate because of the large size of the IMDB dataset (25000 training samples). This method makes sure that performance differences can be attributed to changes in model architecture, capacity, or regularization, rather than differences in the validation examples. Additionally, this makes sure that the test set remains entirely separate for final evaluation, providing an unbiased estimate of real world generalization performance.

To ensure that performance differences are not affected by changes in validation samples, the same validation partition is retained across all runs. Prior to creating the split, the vectorized training data is shuffled once using a fixed random seed. This helps to improves reproducibility and ensures that the validation subset is not affected by any un wanted ordering in the original data.

The 80/20 split is used for the original 25000 training samples, with 5000 samples reserved for validation and the remaining 20000 samples used for training. This is a typical validation split and ensures that a sufficient amount of data is available for both training and validation.

In [94]:
```
# set the number of samples that will be used for the validation dataset[1]
validation_size = 5000

# create an array of indices based on the number of training samples
indices = np.arange(len(x_train_vec))

# set a random seed so the shuffling is reproducible each time the code runs
np.random.seed(42)

# randomly shuffle the indices so the training data is mixed
np.random.shuffle(indices)

# reorder the training data using the shuffled indices
x_train_vec = x_train_vec[indices]
# convert the labels to a numpy array and reorder them using the same shuffled indi
y_train = np.array(y_train)[indices]
# take the first 5000 samples from the shuffled training data to use as the validat
x_val = x_train_vec[:validation_size]
# take the remaining samples after the first 5000 to use as the actual training dat
```

```
partial_x_train = x_train_vec[validation_size:]
# take the first 5000 labels to match the validation data
y_val = y_train[:validation_size]
# take the remaining labels to match the partial training data
partial_y_train = y_train[validation_size:]
```

After the split, the data is organized as follows:

partial_x_train: 20000 training samples

x_val: 5000 validation samples

x_test_vec: 25000 test samples

This approach of the hold out validation ensures that the validation data remains fixed across all the models, allowing for meaningful comparisons between different models and configurations. The goal of this data preparation stage is to ensure controlled experimentation, making sure reliable observations of underfitting, overfitting, and the effects of regularization in later stages of the DLWP 4.5 workflow.

5. Baseline Model

5.1 Objective of the baseline model

The baseline model is a feedforward neural network implemented using the Keras Sequential API. A Sequential model stacks layers in a linear order, with the output of one layer serving as the input to the next. Consistent with the project's constraints to use only DLWP Part 1 components (Chapters 1–4), the baseline model utilizes Dense layers exclusively. Each review is represented as a fixed-length bag-of-words vector ,so the network input shape is (10000)

$$x \in \mathbb{R}^{10000}$$

The purpose of the baseline is to establish a small reference model with statistical power ,one that performs meaningfully better than a simple baseline (≈50% accuracy) and provides a controlled benchmark for later experiments that systematically scale model capacity and apply regularization.

5.2 How the Model works

During a forward pass, the first Dense layer computes a linear transformation of the input vector using a weight matrix and bias term:

$$h = g(W^T x + b)$$

[2]

where : g(z)=max(0,z)[2] is the ReLU activation function, introducing non-linearity to the model. This allows the network to learn non linear relationships between word patterns and sentiment. The final output layer consists of a single unit with a sigmoid activation function, which maps the output to the range [0,1] representing the probability that the review is positive:

$$P(y = 1 \mid x)$$

[2]

5.3 Justification

The baseline architecture uses a single hidden Dense layer with 8 units followed by a one-unit sigmoid output layer. This is intentionally smaller than the reference model in DLWP (which uses two hidden layers of 16 units each) to provide a controlled starting point for future experiments. The aim is to create a model that achieves statistical power, meaningfully outperforming the ~50% baseline for binary classification, while still leaving room for more complex models to overfit as we increase capacity.

The ReLU activation introduces necessary non-linearity, preventing the stacked Dense layers from reducing to a linear transformation, while the sigmoid activation is appropriate for binary classification. The model is trained using binary cross-entropy loss, consistent with its output structure.

5.4 Training Configuration

The model is compiled using binary cross-entropy loss, which is suitable for binary sentiment classification with a single sigmoid output unit. The RMSprop optimizer is used because it provides stable convergence for deep networks without requiring significant hyperparameter tuning.

Performance is monitored using accuracy, which is appropriate for this balanced dataset. In addition to accuracy, loss is tracked throughout training, as it provides earlier insights into overfitting than accuracy alone.

For the initial baseline run, training is performed for 20 epochs using mini-batches of 512 samples. Model performance is evaluated on the fixed hold-out validation set at the end of each epoch. The purpose of this first run is not to assume the best model will occur at the final epoch, but rather to inspect the learning curves and identify the point at which validation performance no longer improves. This process follows the DLWP workflow, where the validation set is used to determine the optimal training duration before any final test evaluation.

In [47]:
```
# import keras from tensorflow, which provides high level tools for building neural
from tensorflow import keras
```

```
# import the layers module so we can add different layers to the neural network
from tensorflow.keras import layers



# create a simple baseline neural network model using the sequential api
baseline_model = keras.Sequential([

    # add a dense (fully connected) layer with 8 neurons and relu activation
    # input_shape=(10000,) means each input review is represented by a vector of si
    layers.Dense(8, activation='relu', input_shape=(10000,)),

    # add the output layer with 1 neuron and sigmoid activation
    # sigmoid outputs a value between 0 and 1 which represents the probability of p
    layers.Dense(1, activation='sigmoid')
])

# configure the model before training
baseline_model.compile(

    # use rmsprop as the optimizer to update the model weights during training
    optimizer='rmsprop',

    # use binary crossentropy because this is a binary classification problem (posi
    loss='binary_crossentropy',

    # track accuracy during training to see how well the model performs
    metrics=['accuracy']
)

# train the baseline model using the training data
history_baseline = baseline_model.fit(

    # the training feature vectors
    partial_x_train,

    # the training labels
    partial_y_train,

    # train the model for 20 complete passes through the training dataset
    epochs=20,

    # process 512 samples at a time during training
    batch_size=512,

    # use the validation dataset to monitor model performance after each epoch
    validation_data=(x_val, y_val)
)
```

```
C:\Users\vinay\anaconda3\Lib\site-packages\keras\src\layers\core\dense.py:106: UserW
arning: Do not pass an `input_shape`/`input_dim` argument to a layer. When using Seq
uential models, prefer using an `Input(shape)` object as the first layer in the mode
l instead.
  super().__init__(activity_regularizer=activity_regularizer, **kwargs)
```

```
Epoch 1/20
40/40 ━━━━━━━━━━━━━━━━━━━━ 4s 47ms/step - accuracy: 0.8008 - loss: 0.5174 - val_accu
racy: 0.8602 - val_loss: 0.4141
Epoch 2/20
40/40 ━━━━━━━━━━━━━━━━━━━━ 2s 26ms/step - accuracy: 0.8895 - loss: 0.3509 - val_accu
racy: 0.8786 - val_loss: 0.3383
Epoch 3/20
40/40 ━━━━━━━━━━━━━━━━━━━━ 2s 31ms/step - accuracy: 0.9115 - loss: 0.2807 - val_accu
racy: 0.8830 - val_loss: 0.3044
Epoch 4/20
40/40 ━━━━━━━━━━━━━━━━━━━━ 3s 36ms/step - accuracy: 0.9212 - loss: 0.2403 - val_accu
racy: 0.8826 - val_loss: 0.2926
Epoch 5/20
40/40 ━━━━━━━━━━━━━━━━━━━━ 1s 25ms/step - accuracy: 0.9314 - loss: 0.2122 - val_accu
racy: 0.8848 - val_loss: 0.2785
Epoch 6/20
40/40 ━━━━━━━━━━━━━━━━━━━━ 2s 48ms/step - accuracy: 0.9366 - loss: 0.1916 - val_accu
racy: 0.8788 - val_loss: 0.2899
Epoch 7/20
40/40 ━━━━━━━━━━━━━━━━━━━━ 2s 33ms/step - accuracy: 0.9427 - loss: 0.1756 - val_accu
racy: 0.8794 - val_loss: 0.2848
Epoch 8/20
40/40 ━━━━━━━━━━━━━━━━━━━━ 3s 33ms/step - accuracy: 0.9466 - loss: 0.1622 - val_accu
racy: 0.8848 - val_loss: 0.2783
Epoch 9/20
40/40 ━━━━━━━━━━━━━━━━━━━━ 2s 27ms/step - accuracy: 0.9519 - loss: 0.1505 - val_accu
racy: 0.8766 - val_loss: 0.2949
Epoch 10/20
40/40 ━━━━━━━━━━━━━━━━━━━━ 2s 35ms/step - accuracy: 0.9560 - loss: 0.1404 - val_accu
racy: 0.8832 - val_loss: 0.2826
Epoch 11/20
40/40 ━━━━━━━━━━━━━━━━━━━━ 1s 23ms/step - accuracy: 0.9589 - loss: 0.1310 - val_accu
racy: 0.8744 - val_loss: 0.3020
Epoch 12/20
40/40 ━━━━━━━━━━━━━━━━━━━━ 1s 27ms/step - accuracy: 0.9615 - loss: 0.1233 - val_accu
racy: 0.8838 - val_loss: 0.2940
Epoch 13/20
40/40 ━━━━━━━━━━━━━━━━━━━━ 1s 28ms/step - accuracy: 0.9653 - loss: 0.1150 - val_accu
racy: 0.8812 - val_loss: 0.3024
Epoch 14/20
40/40 ━━━━━━━━━━━━━━━━━━━━ 1s 27ms/step - accuracy: 0.9676 - loss: 0.1090 - val_accu
racy: 0.8736 - val_loss: 0.3195
Epoch 15/20
40/40 ━━━━━━━━━━━━━━━━━━━━ 2s 42ms/step - accuracy: 0.9694 - loss: 0.1033 - val_accu
racy: 0.8800 - val_loss: 0.3134
Epoch 16/20
40/40 ━━━━━━━━━━━━━━━━━━━━ 1s 24ms/step - accuracy: 0.9722 - loss: 0.0975 - val_accu
racy: 0.8760 - val_loss: 0.3293
Epoch 17/20
40/40 ━━━━━━━━━━━━━━━━━━━━ 2s 28ms/step - accuracy: 0.9737 - loss: 0.0915 - val_accu
racy: 0.8708 - val_loss: 0.3490
Epoch 18/20
40/40 ━━━━━━━━━━━━━━━━━━━━ 2s 58ms/step - accuracy: 0.9765 - loss: 0.0871 - val_accu
racy: 0.8752 - val_loss: 0.3412
Epoch 19/20
40/40 ━━━━━━━━━━━━━━━━━━━━ 2s 25ms/step - accuracy: 0.9776 - loss: 0.0825 - val_accu
```

```
racy: 0.8754 - val_loss: 0.3474
Epoch 20/20
40/40 ━━━━━━━━━━━━━━━━━━━━ 1s 18ms/step - accuracy: 0.9795 - loss: 0.0770 - val_accu
racy: 0.8758 - val_loss: 0.3549
```

In [48]:
```python
# import numpy for numerical operations such as finding maximum, minimum, and index
import numpy as np

# find the highest validation accuracy achieved during training
best_val_acc = max(history_baseline.history['val_accuracy'])

# find the lowest validation loss achieved during training
best_val_loss = min(history_baseline.history['val_loss'])

# find the epoch number where the highest validation accuracy occurred
# np.argmax returns the index, so we add 1 because epochs start from 1
best_epoch_val_acc = np.argmax(history_baseline.history['val_accuracy']) + 1

# find the epoch number where the lowest validation loss occurred
# np.argmin returns the index of the smallest value
best_epoch_val_loss = np.argmin(history_baseline.history['val_loss']) + 1

# get the final training accuracy from the last epoch
final_train_acc = history_baseline.history['accuracy'][-1]

# get the final training loss from the last epoch
final_train_loss = history_baseline.history['loss'][-1]

# print the best validation accuracy achieved by the model
print("Best validation accuracy:", best_val_acc)

# print the lowest validation loss achieved by the model
print("Best validation loss:", best_val_loss)

# print the epoch number where the best validation accuracy occurred
print("Best epoch by validation accuracy:", best_epoch_val_acc)

# print the epoch number where the best validation loss occurred
print("Best epoch by validation loss:", best_epoch_val_loss)

# print the final training accuracy after the last epoch
print("Final training accuracy:", final_train_acc)

# print the final training loss after the last epoch
print("Final training loss:", final_train_loss)
```

```
Best validation accuracy: 0.8848000168800354
Best validation loss: 0.2782781720161438
Best epoch by validation accuracy: 5
Best epoch by validation loss: 8
Final training accuracy: 0.9794999957084656
Final training loss: 0.0770399272441864
```

In [102…
```python
baseline_model.summary()
```

```
Model: "sequential_5"
```

Layer (type)	Output Shape	Param #
dense_15 (Dense)	(None, 8)	80,008
dense_16 (Dense)	(None, 1)	9

Total params: 160,036 (625.14 KB)
Trainable params: 80,017 (312.57 KB)
Non-trainable params: 0 (0.00 B)
Optimizer params: 80,019 (312.58 KB)

5.5 Results and Interpretation

The baseline model consists of a single hidden Dense layer with 8 ReLU units, followed by a single sigmoid output unit. I kept this architecture simple so that it would act as a low capacity reference point for later experiments. Its role is to provide a clear benchmark, allowing changes in model size, regularization, and training strategy to be evaluated objectively.

Since the IMDB dataset is balanced between positive and negative reviews, a naive classifier would achieve roughly 50% accuracy by guessing randomly. This baseline model performs far above that level. Even in the first epoch, validation accuracy reached 0.8602, showing that the model quickly begins learning meaningful sentiment patterns from the vectorized reviews rather than making random predictions.

Training performance improved steadily across epochs. Validation accuracy increased from 0.8602 in epoch 1 to a maximum of 0.8848, first achieved at epoch 5 (and matched again at epoch 8). Validation loss continued decreasing slightly beyond this point, reaching the minimum value of 0.2783 at epoch 8. Although epoch 5 gives the highest validation accuracy, epoch 8 provides the lowest validation loss. Since validation loss is a more sensitive indicator of generalization performance, epoch 8 can be considered the most technically appropriate stopping point.

Training accuracy continues to increase consistently, reaching 0.9795 by epoch 20, while training loss falls to 0.0770. In contrast, validation performance no longer improves. Validation accuracy changes and gradually decreases from its peak of 0.8848 to 0.8758 by epoch 20, while validation loss increases significantly from 0.2783 to 0.3549. This widening gap between training and validation metrics indicates that the model is increasingly fitting the training data more closely without improving its ability to generalize to unseen reviews.

The increasing gap between training and validation performance highlights that the model is overfitting, suggesting that regularization strategies (dropout or L2 regularization) may be necessary in future experiments to control overfitting and improve generalization.

Overall, the baseline model performs strongly as an initial benchmark and shows the expected behavior described in deep learning theory. Even a simple fully connected network can achieve high validation performance on the vectorized IMDB dataset. However, continued training beyond the optimal epoch leads to overfitting. This confirms that the baseline model provides a reliable reference point for evaluating whether architectural or regularization changes can improve generalization rather than simply increasing training accuracy.

Learning Curve Analysis and Interpretation

In [52]:
```
# import matplotlib so we can create graphs to visualize the training results
import matplotlib.pyplot as plt

# get the training loss values recorded after each epoch
loss = history_baseline.history['loss']

# get the validation loss values recorded after each epoch
val_loss = history_baseline.history['val_loss']

# create a range of epoch numbers starting from 1 up to the total number of epochs
epochs = range(1, len(loss) + 1)

# create a new figure and set its size so the graph is easier to read
plt.figure(figsize=(8,5))

# plot the training loss against the epochs
plt.plot(epochs, loss, label='Training Loss')

# plot the validation loss against the epochs for comparison
plt.plot(epochs, val_loss, label='Validation Loss')

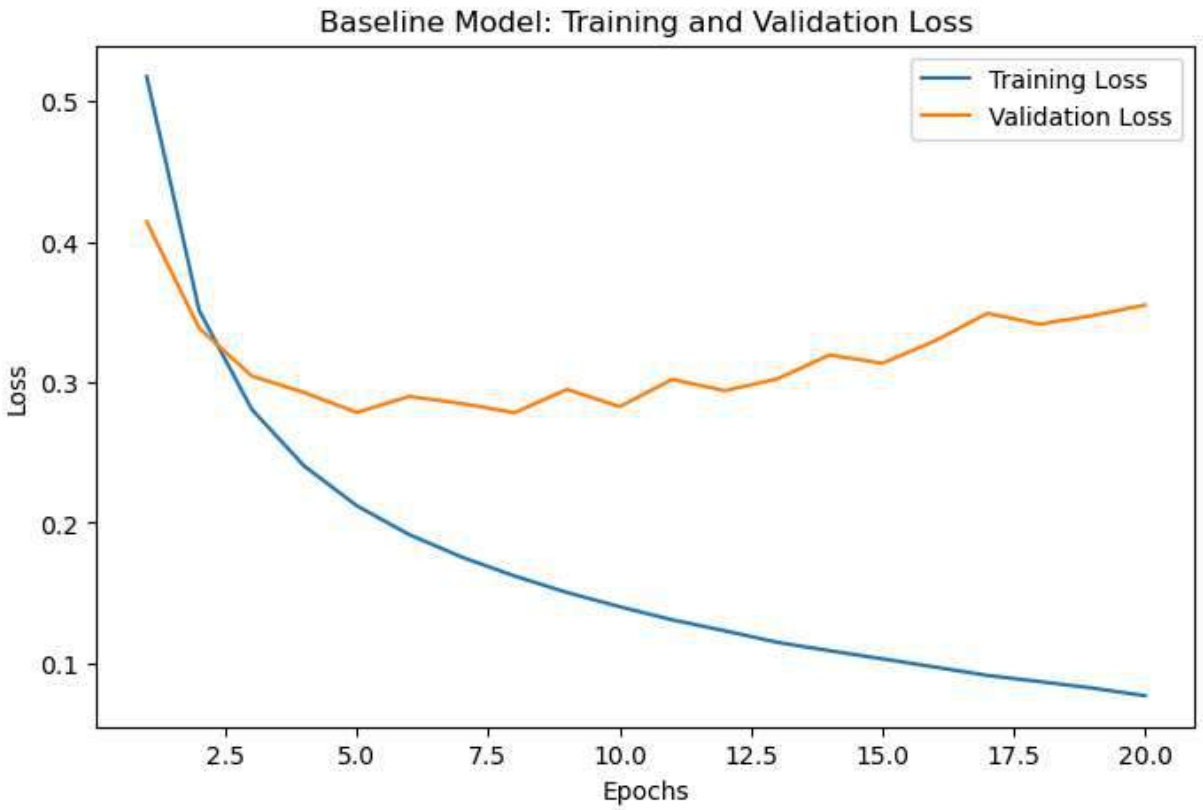
# label the x axis to indicate it represents the epochs
plt.xlabel('Epochs')

# label the y axis to indicate it represents the loss value
plt.ylabel('Loss')

# add a title
plt.title('Baseline Model: Training and Validation Loss')

# show a legend
plt.legend()

# display the graph
plt.show()
```



In [53]:
```
# get the training accuracy values recorded after each epoch
acc = history_baseline.history['accuracy']

# get the validation accuracy values recorded after each epoch
val_acc = history_baseline.history['val_accuracy']

# create a range of epoch numbers starting from 1 up to the total number of epochs
epochs = range(1, len(acc) + 1)

# create a new figure and set its size so the graph is easier to read
plt.figure(figsize=(8,5))

# plot the training accuracy against the epochs
plt.plot(epochs, acc, label='Training Accuracy')

# plot the validation accuracy against the epochs for comparison
plt.plot(epochs, val_acc, label='Validation Accuracy')

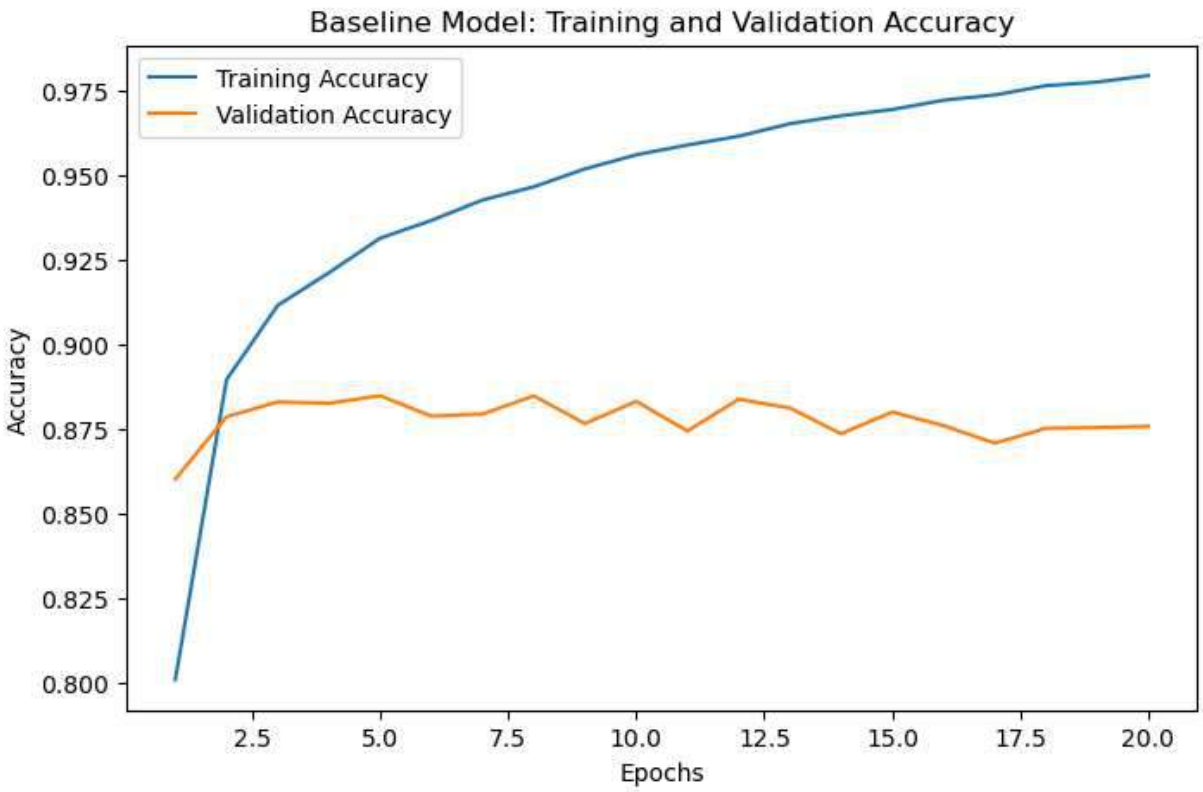
# label the x axis to indicate it represents the epochs
plt.xlabel('Epochs')

# label the y axis to indicate it represents the accuracy values
plt.ylabel('Accuracy')

# add  title
plt.title('Baseline Model: Training and Validation Accuracy')

# show a legend
plt.legend()
```

```
# display the graph
plt.show()
```



Training and Validation Loss:

Initially, both training loss and validation loss decrease sharply, which indicates that the model is learning useful features in the data. This is a good sign, especially in the early epochs, where the model starts recognizing patterns related to sentiment.

Validation loss reaches its minimum of 0.2783 at epoch 8, after which it begins to rise, suggesting that the model is starting to overfit the training data. This is consistent with the concept of generalization: as the model becomes more specialized in the training data, it starts to lose its ability to generalize to new, unseen data.

The sharp decrease in training loss continues as expected, reaching 0.0770 by the final epoch, showing that the model is becoming increasingly accurate in fitting the training data.

Training and Validation Accuracy:

Training accuracy improves steadily, reaching 0.9795 by epoch 20. This is expected as the model becomes increasingly confident in its predictions on the training data.

Validation accuracy, on the other hand, peaks at 0.8848 around epochs 5 to 8, which is where the model shows its best ability to generalize to unseen data.

After epoch 8, validation accuracy fluctuates and gradually declines to 0.8758 by epoch 20. This decline is a direct result of the model overfitting, where it begins to focus too heavily on

memorizing the training data instead of maintaining generalization to unseen data.

Expected Behavior of a Small Model:

The behavior observed in the learning curves is expected for a small neural network trained without regularization. Smaller networks tend to have limited capacity to generalize, which leads to overfitting when training continues past the optimal point.

Test Set Evaluation and Results Interpretation

In [55]:
```
# create a new sequential model that will be trained on the full training dataset
final_baseline_model = keras.Sequential([

    # add a dense layer with 8 neurons and relu activation to learn patterns from t
    # input_shape=(10000,) means each input review is represented by a vector of 10
    layers.Dense(8, activation='relu', input_shape=(10000,)),

    # add the output layer with 1 neuron and sigmoid activation for binary classifi
    # this outputs a probability between 0 and 1 for positive or negative sentiment
    layers.Dense(1, activation='sigmoid')
])

# configure the model before training
final_baseline_model.compile(


    optimizer='rmsprop',


    loss='binary_crossentropy',

    metrics=['accuracy']
)

# train the final model using the full training dataset
final_baseline_model.fit(

    x_train_vec,

    y_train,

    epochs=20,

    batch_size=512
)

# evaluate the trained model using the test dataset to measure its performance on u
test_loss, test_acc = final_baseline_model.evaluate(x_test_vec, y_test)

print("Test loss:", test_loss)

print("Test accuracy:", test_acc)
```

```
Epoch 1/20
49/49 ━━━━━━━━━━━━━━━━━━━━ 3s 23ms/step - accuracy: 0.8095 - loss: 0.4914
Epoch 2/20
49/49 ━━━━━━━━━━━━━━━━━━━━ 1s 19ms/step - accuracy: 0.8974 - loss: 0.3152
Epoch 3/20
49/49 ━━━━━━━━━━━━━━━━━━━━ 1s 24ms/step - accuracy: 0.9158 - loss: 0.2527
Epoch 4/20
49/49 ━━━━━━━━━━━━━━━━━━━━ 2s 27ms/step - accuracy: 0.9274 - loss: 0.2189
Epoch 5/20
49/49 ━━━━━━━━━━━━━━━━━━━━ 1s 23ms/step - accuracy: 0.9334 - loss: 0.1960
Epoch 6/20
49/49 ━━━━━━━━━━━━━━━━━━━━ 2s 29ms/step - accuracy: 0.9397 - loss: 0.1791
Epoch 7/20
49/49 ━━━━━━━━━━━━━━━━━━━━ 1s 23ms/step - accuracy: 0.9440 - loss: 0.1658
Epoch 8/20
49/49 ━━━━━━━━━━━━━━━━━━━━ 2s 31ms/step - accuracy: 0.9486 - loss: 0.1542
Epoch 9/20
49/49 ━━━━━━━━━━━━━━━━━━━━ 2s 14ms/step - accuracy: 0.9527 - loss: 0.1445
Epoch 10/20
49/49 ━━━━━━━━━━━━━━━━━━━━ 1s 16ms/step - accuracy: 0.9559 - loss: 0.1357
Epoch 11/20
49/49 ━━━━━━━━━━━━━━━━━━━━ 2s 22ms/step - accuracy: 0.9595 - loss: 0.1283
Epoch 12/20
49/49 ━━━━━━━━━━━━━━━━━━━━ 1s 25ms/step - accuracy: 0.9612 - loss: 0.1222
Epoch 13/20
49/49 ━━━━━━━━━━━━━━━━━━━━ 2s 19ms/step - accuracy: 0.9653 - loss: 0.1148
Epoch 14/20
49/49 ━━━━━━━━━━━━━━━━━━━━ 1s 19ms/step - accuracy: 0.9671 - loss: 0.1093
Epoch 15/20
49/49 ━━━━━━━━━━━━━━━━━━━━ 2s 29ms/step - accuracy: 0.9681 - loss: 0.1039
Epoch 16/20
49/49 ━━━━━━━━━━━━━━━━━━━━ 1s 18ms/step - accuracy: 0.9689 - loss: 0.0995
Epoch 17/20
49/49 ━━━━━━━━━━━━━━━━━━━━ 1s 22ms/step - accuracy: 0.9723 - loss: 0.0934
Epoch 18/20
49/49 ━━━━━━━━━━━━━━━━━━━━ 2s 30ms/step - accuracy: 0.9738 - loss: 0.0897
Epoch 19/20
49/49 ━━━━━━━━━━━━━━━━━━━━ 3s 37ms/step - accuracy: 0.9762 - loss: 0.0845
Epoch 20/20
49/49 ━━━━━━━━━━━━━━━━━━━━ 2s 33ms/step - accuracy: 0.9775 - loss: 0.0811
782/782 ━━━━━━━━━━━━━━━━━━━━ 6s 7ms/step - accuracy: 0.8679 - loss: 0.3913
Test loss: 0.39134588837623596
Test accuracy: 0.8678799867630005
```

The baseline model is retrained on the full 25000 sample training set for 20 epochs and then evaluated on the held out test set. After retraining, the model achieves a test loss of 0.3913 and a test accuracy of 0.8679.

These results are slightly lower than the best validation performance observed earlier, where validation accuracy peaked at 0.8848 (epoch 5) and validation loss reached a minimum of 0.2783 (epoch 8). This difference is expected and consistent with the overfitting pattern observed during training. After the optimal validation epoch, training accuracy continued to rise steadily, reaching 0.9775 by epoch 20, while validation performance began to change and gradually decrease.

Overall, while the retrained models performance on the test set is slightly worse than the best validation performance, it still demonstrates that the baseline model can achieve a high test accuracy of 86.79%. This confirms that the baseline provides a solid reference point, and future models with increased capacity or regularization can improve on this generalization.

6. Developing Overfitting Model

6.1 Objective of Overfitting Model

The overfitting model increases representational capacity relative to the baseline by increasing both width and depth. Specifically, it uses two hidden Dense layers with 256 units each, compared with the baseline models single hidden layer of 8 units. This increase in the number of trainable parameters allows the model to fit the training data more closely, but also increases the risk of overfitting.

The choice of 256 units per hidden layer is intentional. With a 10000 dimensional input space, a hidden layer of this size provides the network with enough capacity to learn a much larger number of feature combinations and interactions than the baseline model. This makes it easier for the model to represent highly specific patterns in the training reviews, including patterns that may not generalize well to unseen data. The increased capacity is large enough to create a clear contrast with the baseline, while still being within the scope of the coursework.

Adding a second hidden layer increases model capacity further by allowing the network to learn more complex representations. While a single hidden layer can already model useful sentiment patterns, a second layer enables the model to combine lower level features into more specialized higher level patterns. This additional depth increases the models expressiveness, but it also increases the risk that it will overfit the training data by memorizing highly specific patterns that do not generalize.

6.2 Architecture Design and Justification

In this overfitting model, two hidden Dense layers with 256 units each are followed by a single sigmoid output unit. The choice of ReLU activation for the hidden layers ensures that the network can learn non linear relationships between patterns in the data. ReLU activation introduces non linearity, which prevents the network from becoming a series of linear transformations, thus allowing the model to capture complex patterns in the data.

While increasing the width (number of units) and depth (number of layers) of the model allows the network to learn richer feature combinations, it also increases the risk of overfitting. This experiment is designed to reveal how increasing model capacity leads to

overfitting as the model becomes more specialized to the training data, fitting noise or patterns that do not generalize well to new unseen data.

6.3 Training Configuration

The overfitting model is trained for 30 epochs, which is longer than the baselines 20 epoch training period. This extended training duration allows the high capacity model to fit the training data more closely, and the learning curves will clearly reveal the divergence between training and validation performance.

The batch size of 512 is consistent with the baseline, ensuring fair comparisons between the models and maintaining computational efficiency. The batch size also ensures stable gradient estimates, which is particularly important when training with high dimensional sparse input vectors.

As with the baseline model, the overfitting model is compiled using binary cross entropy loss, appropriate for the binary classification task, and RMSprop optimizer is retained to provide stable optimization.

6.4 Justification

By keeping the optimizer and loss function consistent with the baseline model, any observed changes in performance can be confidently attributed to the increased capacity of the model. The aim of this experiment is to explore the effects of increased model capacity and the tendency for models with high capacity to overfit the training data. This configuration will provide insights into how the model's complexity impacts its generalization ability, revealing the overfitting pattern when the model is allowed to fit the training data too closely.

6.4 Overfitting Model

In [66]:
```
from tensorflow import keras
from tensorflow.keras import layers

# create a larger neural network with more neurons, which can lead to overfitting
overfit_model = keras.Sequential([
    layers.Dense(256, activation="relu", input_shape=(10000,)),
    layers.Dense(256, activation="relu"),  # second hidden layer with many neurons
    layers.Dense(1, activation="sigmoid")  # output layer for binary classification
])

# compile the model by defining the optimizer, loss function, and evaluation metric
overfit_model.compile(
    optimizer="rmsprop",
    loss="binary_crossentropy",
    metrics=["accuracy"]
)
```

```
# train the model and store the training history
history_overfit = overfit_model.fit(
    partial_x_train,
    partial_y_train,
    epochs=30,  # more epochs increase the chance of overfitting
    batch_size=512,
    validation_data=(x_val, y_val)  # validation data used to monitor performance
)

# store the key results from this experiment for later comparison
results.append([
    "Overfit (256,256)",
    max(history_overfit.history["val_accuracy"]),  # best validation accuracy achie
    min(history_overfit.history["val_loss"]),
    history_overfit.history["accuracy"][-1],  # final training accuracy
    history_overfit.history["loss"][-1]
])
```

```
Epoch 1/30
40/40 ━━━━━━━━━━━━━━━━━━━━ 10s 190ms/step - accuracy: 0.7421 - loss: 0.5119 - val_ac
curacy: 0.8560 - val_loss: 0.3588
Epoch 2/30
40/40 ━━━━━━━━━━━━━━━━━━━━ 9s 221ms/step - accuracy: 0.8853 - loss: 0.2824 - val_acc
uracy: 0.8572 - val_loss: 0.3446
Epoch 3/30
40/40 ━━━━━━━━━━━━━━━━━━━━ 11s 237ms/step - accuracy: 0.9117 - loss: 0.2226 - val_ac
curacy: 0.8838 - val_loss: 0.2729
Epoch 4/30
40/40 ━━━━━━━━━━━━━━━━━━━━ 8s 210ms/step - accuracy: 0.9319 - loss: 0.1736 - val_acc
uracy: 0.8828 - val_loss: 0.3038
Epoch 5/30
40/40 ━━━━━━━━━━━━━━━━━━━━ 8s 190ms/step - accuracy: 0.9477 - loss: 0.1381 - val_acc
uracy: 0.8558 - val_loss: 0.3749
Epoch 6/30
40/40 ━━━━━━━━━━━━━━━━━━━━ 9s 153ms/step - accuracy: 0.9617 - loss: 0.1043 - val_acc
uracy: 0.8762 - val_loss: 0.3288
Epoch 7/30
40/40 ━━━━━━━━━━━━━━━━━━━━ 6s 138ms/step - accuracy: 0.9704 - loss: 0.0793 - val_acc
uracy: 0.8724 - val_loss: 0.3436
Epoch 8/30
40/40 ━━━━━━━━━━━━━━━━━━━━ 7s 162ms/step - accuracy: 0.9830 - loss: 0.0620 - val_acc
uracy: 0.8800 - val_loss: 0.3717
Epoch 9/30
40/40 ━━━━━━━━━━━━━━━━━━━━ 5s 128ms/step - accuracy: 0.9857 - loss: 0.0483 - val_acc
uracy: 0.8792 - val_loss: 0.3503
Epoch 10/30
40/40 ━━━━━━━━━━━━━━━━━━━━ 6s 150ms/step - accuracy: 0.9991 - loss: 0.0098 - val_acc
uracy: 0.8794 - val_loss: 0.5090
Epoch 11/30
40/40 ━━━━━━━━━━━━━━━━━━━━ 6s 134ms/step - accuracy: 0.9873 - loss: 0.0545 - val_acc
uracy: 0.8798 - val_loss: 0.4427
Epoch 12/30
40/40 ━━━━━━━━━━━━━━━━━━━━ 11s 164ms/step - accuracy: 0.9998 - loss: 0.0036 - val_ac
curacy: 0.8704 - val_loss: 0.6381
Epoch 13/30
40/40 ━━━━━━━━━━━━━━━━━━━━ 6s 137ms/step - accuracy: 0.9862 - loss: 0.0564 - val_acc
uracy: 0.8808 - val_loss: 0.5007
Epoch 14/30
40/40 ━━━━━━━━━━━━━━━━━━━━ 7s 172ms/step - accuracy: 1.0000 - loss: 0.0016 - val_acc
uracy: 0.8768 - val_loss: 0.6125
Epoch 15/30
40/40 ━━━━━━━━━━━━━━━━━━━━ 9s 131ms/step - accuracy: 0.9883 - loss: 0.0556 - val_acc
uracy: 0.8770 - val_loss: 0.4819
Epoch 16/30
40/40 ━━━━━━━━━━━━━━━━━━━━ 5s 130ms/step - accuracy: 1.0000 - loss: 0.0018 - val_acc
uracy: 0.8786 - val_loss: 0.5573
Epoch 17/30
40/40 ━━━━━━━━━━━━━━━━━━━━ 7s 162ms/step - accuracy: 1.0000 - loss: 7.2080e-04 - val
_accuracy: 0.8768 - val_loss: 0.6359
Epoch 18/30
40/40 ━━━━━━━━━━━━━━━━━━━━ 10s 144ms/step - accuracy: 1.0000 - loss: 4.2138e-04 - va
l_accuracy: 0.8778 - val_loss: 0.6818
Epoch 19/30
40/40 ━━━━━━━━━━━━━━━━━━━━ 10s 139ms/step - accuracy: 1.0000 - loss: 2.7674e-04 - va
```

```
l_accuracy: 0.8766 - val_loss: 0.7121
Epoch 20/30
40/40 ━━━━━━━━━━━━━━━━━━━━ 6s 137ms/step - accuracy: 1.0000 - loss: 2.0201e-04 - val
_accuracy: 0.8794 - val_loss: 0.7345
Epoch 21/30
40/40 ━━━━━━━━━━━━━━━━━━━━ 6s 151ms/step - accuracy: 1.0000 - loss: 1.6001e-04 - val
_accuracy: 0.8798 - val_loss: 0.7511
Epoch 22/30
40/40 ━━━━━━━━━━━━━━━━━━━━ 10s 133ms/step - accuracy: 1.0000 - loss: 1.3162e-04 - va
l_accuracy: 0.8798 - val_loss: 0.7678
Epoch 23/30
40/40 ━━━━━━━━━━━━━━━━━━━━ 7s 172ms/step - accuracy: 1.0000 - loss: 1.1123e-04 - val
_accuracy: 0.8786 - val_loss: 0.7815
Epoch 24/30
40/40 ━━━━━━━━━━━━━━━━━━━━ 7s 158ms/step - accuracy: 1.0000 - loss: 9.7197e-05 - val
_accuracy: 0.8786 - val_loss: 0.7923
Epoch 25/30
40/40 ━━━━━━━━━━━━━━━━━━━━ 6s 154ms/step - accuracy: 1.0000 - loss: 8.5438e-05 - val
_accuracy: 0.8772 - val_loss: 0.8105
Epoch 26/30
40/40 ━━━━━━━━━━━━━━━━━━━━ 10s 137ms/step - accuracy: 1.0000 - loss: 7.7110e-05 - va
l_accuracy: 0.8764 - val_loss: 0.8152
Epoch 27/30
40/40 ━━━━━━━━━━━━━━━━━━━━ 6s 147ms/step - accuracy: 1.0000 - loss: 6.8984e-05 - val
_accuracy: 0.8776 - val_loss: 0.8266
Epoch 28/30
40/40 ━━━━━━━━━━━━━━━━━━━━ 10s 139ms/step - accuracy: 1.0000 - loss: 6.3373e-05 - va
l_accuracy: 0.8774 - val_loss: 0.8275
Epoch 29/30
40/40 ━━━━━━━━━━━━━━━━━━━━ 11s 153ms/step - accuracy: 1.0000 - loss: 5.7758e-05 - va
l_accuracy: 0.8762 - val_loss: 0.8368
Epoch 30/30
40/40 ━━━━━━━━━━━━━━━━━━━━ 6s 135ms/step - accuracy: 1.0000 - loss: 5.3435e-05 - val
_accuracy: 0.8772 - val_loss: 0.8417
```

In [67]:

```python
import numpy as np

# find the highest validation accuracy achieved during training
best_val_acc_overfit = max(history_overfit.history['val_accuracy'])

# find the lowest validation loss during training
best_val_loss_overfit = min(history_overfit.history['val_loss'])

# get the epoch where the highest validation accuracy occurred
best_epoch_val_acc_overfit = np.argmax(history_overfit.history['val_accuracy']) + 1

# get the epoch where the lowest validation loss occurred
best_epoch_val_loss_overfit = np.argmin(history_overfit.history['val_loss']) + 1

# get the final training accuracy from the last epoch
final_train_acc_overfit = history_overfit.history['accuracy'][-1]

# get the final training loss from the last epoch
final_train_loss_overfit = history_overfit.history['loss'][-1]

print("Best validation accuracy:", best_val_acc_overfit)
```

```
print("Best validation loss:", best_val_loss_overfit)
print("Best epoch by validation accuracy:", best_epoch_val_acc_overfit)
print("Best epoch by validation loss:", best_epoch_val_loss_overfit)
print("Final training accuracy:", final_train_acc_overfit)
print("Final training loss:", final_train_loss_overfit)
```

```
Best validation accuracy: 0.8838000297546387
Best validation loss: 0.2728777229785919
Best epoch by validation accuracy: 3
Best epoch by validation loss: 3
Final training accuracy: 1.0
Final training loss: 5.343545853975229e-05
```

In [99]:
```
overfit_model.summary()
```

Model: "sequential_4"

Layer (type)	Output Shape	Param #
dense_12 (Dense)	(None, 256)	2,560,256
dense_13 (Dense)	(None, 256)	65,792
dense_14 (Dense)	(None, 1)	257

Total params: 5,252,612 (20.04 MB)

Trainable params: 2,626,305 (10.02 MB)

Non-trainable params: 0 (0.00 B)

Optimizer params: 2,626,307 (10.02 MB)

6.5 Results and Interpretation

The overfitting model uses two hidden Dense layers with 256 ReLU units each, making it larger than the baseline model. I increased the models capacity to observe how excessive flexibility affects training behavior and generalization. Compared to the baseline models single 8 unit hidden layer, this configuration contains more trainable parameters, allowing for greater representational power and the ability to memorize patterns in the training data.

During the early epochs, the model improves rapidly. Validation accuracy rises from 0.8560 in epoch 1 to a maximum of 0.8838 at epoch 3. Similarly, validation loss decreases sharply, reaching its minimum value of 0.2729 at epoch 3. Both the best validation accuracy and the lowest validation loss occur at epoch 3, which indicates that the model achieves its optimal generalization performance early in training. After epoch 3, validation performance does not meaningfully improve.

Beyond this early peak, a strong pattern of overfitting becomes evident. Training accuracy continues to increase aggressively, eventually reaching 1.0000, while training loss falls to approximately 0.000053 by epoch 30. This indicates near-perfect memorization of the training data. In contrast, validation accuracy remains relatively stable in the high 0.87 0.88

range and does not exceed the early peak of 0.8838. More significantly, validation loss increases sharply from 0.2729 at epoch 3 to 0.8417 by epoch 30. This dramatic rise in validation loss shows that the model becomes increasingly overconfident in its predictions without improving its ability to generalize to unseen data.

Despite the increase in model capacity, this overfitting model does not outperform the baseline model in terms of validation performance. The baseline model achieved a slightly higher best validation accuracy (0.8848) and a similar best validation loss (0.2783), despite having far fewer parameters. The larger overfitting model memorizes the training data completely, but fails to leverage its capacity for better generalization.

Overall, this model clearly demonstrates the effect of excessive capacity within the deep learning workflow. While it achieves perfect training accuracy, its validation loss deteriorates significantly after the early optimal epoch. This confirms that simply increasing network size does not guarantee better performance and strongly motivates the need for regularization in subsequent experiments.

Learning Curve Analysis and Interpretation

In [71]:
```
# get the training loss values from the overfitting model history
loss = history_overfit.history['loss']

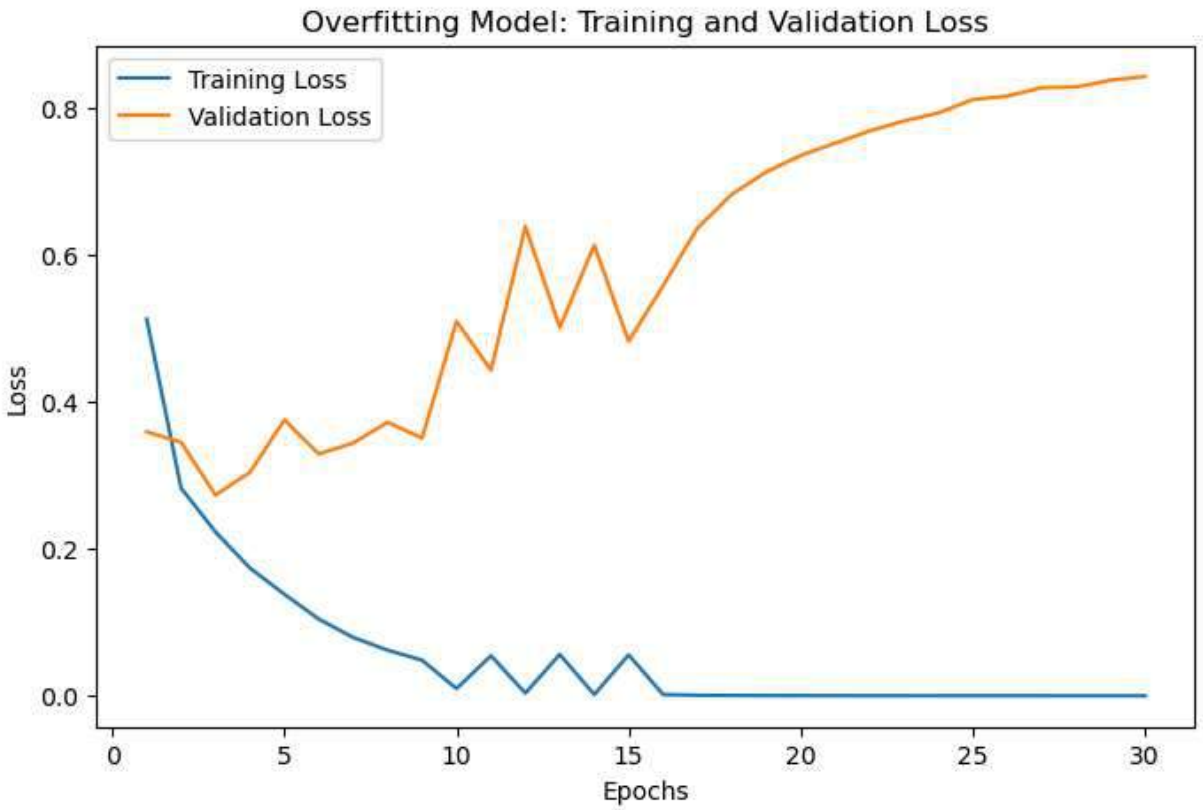
# get the validation loss values to compare with training loss
val_loss = history_overfit.history['val_loss']

# create a list of epoch numbers based on how many epochs were trained
epochs = range(1, len(loss) + 1)

plt.figure(figsize=(8,5))

# plot how the training loss changed over the epochs
plt.plot(epochs, loss, label='Training Loss')

# plot the validation loss to see if the model started overfitting
plt.plot(epochs, val_loss, label='Validation Loss')
plt.xlabel('Epochs')
plt.ylabel('Loss')
plt.title('Overfitting Model: Training and Validation Loss')
plt.legend()
plt.show()
```



In [72]:
```
# get the training accuracy values recorded after each epoch
acc = history_overfit.history['accuracy']

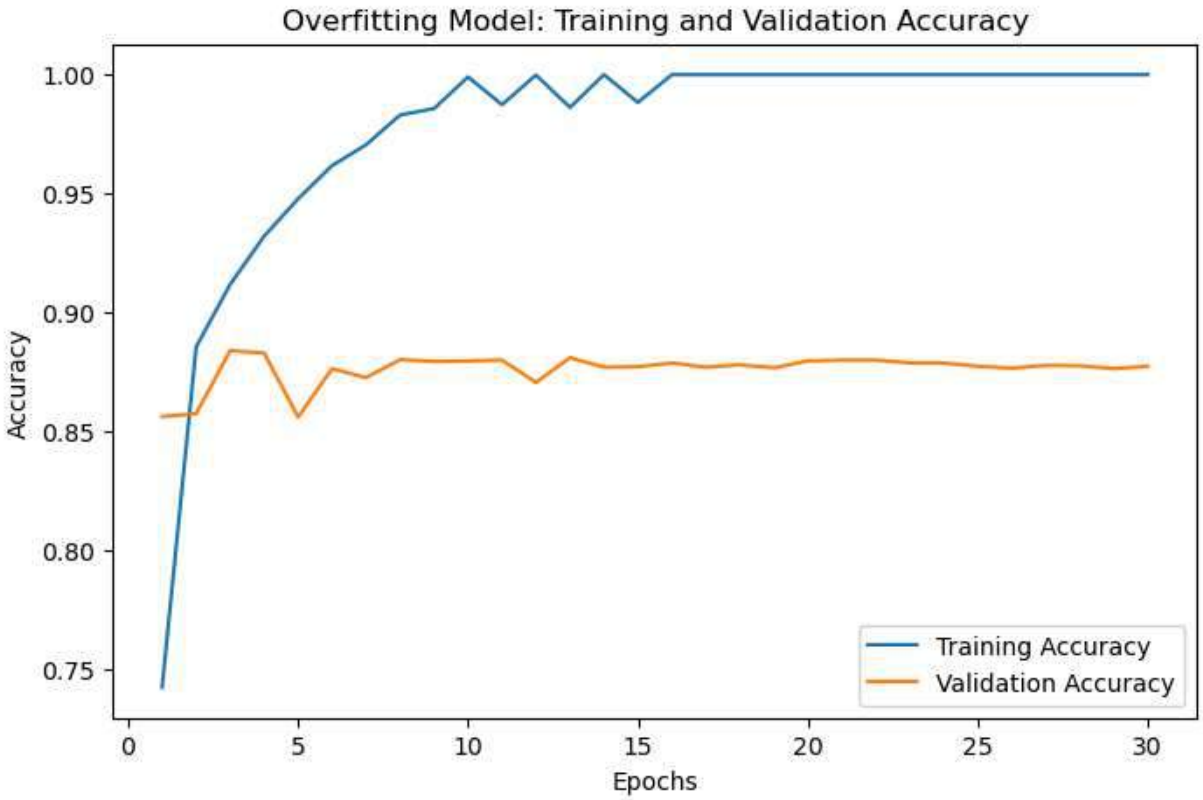
# get the validation accuracy values for comparison
val_acc = history_overfit.history['val_accuracy']

epochs = range(1, len(acc) + 1)

plt.figure(figsize=(8,5))

# plot the training accuracy across epochs
plt.plot(epochs, acc, label='Training Accuracy')

# plot the validation accuracy to compare model performance
plt.plot(epochs, val_acc, label='Validation Accuracy')
plt.xlabel('Epochs')
plt.ylabel('Accuracy')
plt.title('Overfitting Model: Training and Validation Accuracy')
plt.legend()
plt.show()
```



The learning curves for the overfitting model show a dramatic pattern. In the early epochs, both training loss and validation loss decrease rapidly, indicating that the model is learning useful features from the data. Validation loss reaches its minimum of 0.2729 at epoch 3, and validation accuracy peaks at 0.8838 at the same epoch. This suggests that, despite its large capacity, the model is still able to generalize well early on.

However, after epoch 3, the behavior changes sharply. Training loss continues to decrease aggressively, eventually approaching zero (approximately 0.00005), while training accuracy reaches 1.0000. This indicates that the model is memorizing the training data almost perfectly.

In contrast, validation loss increases steadily and dramatically after the early peak. It rises from 0.2729 at epoch 3 to 0.8417 by epoch 30, and validation accuracy remains relatively stable in the 0.87–0.88 range, never surpassing the early peak. The widening gap between near-zero training loss and the rapidly increasing validation loss is clear evidence of overfitting. The model becomes more confident in its predictions on the training data, but fails to improve its ability to generalize to the unseen validation set.

Compared to the baseline model, the divergence between training and validation curves is much larger and occurs much earlier. The overfitting model becomes highly confident on the training data, while its performance on unseen data deteriorates. The additional layers and units in this model allow it to memorize the training data, but they do not improve its ability to generalize.

Overall, this model clearly demonstrates the effect of excessive model capacity in deep learning. Although it achieves perfect training accuracy, its validation loss deteriorates significantly after the early optimal epoch. This confirms that increasing network size without regularization leads to memorization rather than generalization. Techniques such as regularization (e.g., dropout or L2 regularization) would be required to mitigate overfitting and improve the model's ability to generalize to unseen data.

Test Set Evaluation and Results Interpretation

In [75]:
```
# create the final overfitting model with two large hidden layers
final_overfit_model = keras.Sequential([
    layers.Dense(256, activation='relu', input_shape=(10000,)),
    layers.Dense(256, activation='relu'),  # large layers increase model capacity
    layers.Dense(1, activation='sigmoid')  # output layer for binary sentiment pred
])

# compile the model by specifying the optimizer, loss function, and metric
final_overfit_model.compile(
    optimizer='rmsprop',
    loss='binary_crossentropy',
    metrics=['accuracy']
)

# train the model using the full training dataset
final_overfit_model.fit(
    x_train_vec,
    y_train,
    epochs=30,  # more epochs may increase overfitting
    batch_size=512
)

# evaluate the trained model on the unseen test dataset
test_loss_overfit, test_acc_overfit = final_overfit_model.evaluate(x_test_vec, y_te

print("Test loss:", test_loss_overfit)
print("Test accuracy:", test_acc_overfit)
```

```
Epoch 1/30
49/49 ━━━━━━━━━━━━━━━━━━━━ 6s 71ms/step - accuracy: 0.7752 - loss: 0.4743
Epoch 2/30
49/49 ━━━━━━━━━━━━━━━━━━━━ 4s 72ms/step - accuracy: 0.8936 - loss: 0.2646
Epoch 3/30
49/49 ━━━━━━━━━━━━━━━━━━━━ 4s 84ms/step - accuracy: 0.9163 - loss: 0.2136
Epoch 4/30
49/49 ━━━━━━━━━━━━━━━━━━━━ 4s 81ms/step - accuracy: 0.9359 - loss: 0.1700
Epoch 5/30
49/49 ━━━━━━━━━━━━━━━━━━━━ 4s 81ms/step - accuracy: 0.9537 - loss: 0.1254
Epoch 6/30
49/49 ━━━━━━━━━━━━━━━━━━━━ 4s 78ms/step - accuracy: 0.9646 - loss: 0.0998
Epoch 7/30
49/49 ━━━━━━━━━━━━━━━━━━━━ 4s 78ms/step - accuracy: 0.9761 - loss: 0.0739
Epoch 8/30
49/49 ━━━━━━━━━━━━━━━━━━━━ 4s 77ms/step - accuracy: 0.9867 - loss: 0.0482
Epoch 9/30
49/49 ━━━━━━━━━━━━━━━━━━━━ 4s 80ms/step - accuracy: 0.9894 - loss: 0.0448
Epoch 10/30
49/49 ━━━━━━━━━━━━━━━━━━━━ 4s 78ms/step - accuracy: 0.9907 - loss: 0.0372
Epoch 11/30
49/49 ━━━━━━━━━━━━━━━━━━━━ 4s 79ms/step - accuracy: 0.9998 - loss: 0.0041
Epoch 12/30
49/49 ━━━━━━━━━━━━━━━━━━━━ 4s 76ms/step - accuracy: 0.9912 - loss: 0.0424
Epoch 13/30
49/49 ━━━━━━━━━━━━━━━━━━━━ 4s 82ms/step - accuracy: 0.9901 - loss: 0.0484
Epoch 14/30
49/49 ━━━━━━━━━━━━━━━━━━━━ 4s 80ms/step - accuracy: 1.0000 - loss: 0.0021
Epoch 15/30
49/49 ━━━━━━━━━━━━━━━━━━━━ 4s 85ms/step - accuracy: 0.9917 - loss: 0.0391
Epoch 16/30
49/49 ━━━━━━━━━━━━━━━━━━━━ 4s 80ms/step - accuracy: 1.0000 - loss: 0.0016
Epoch 17/30
49/49 ━━━━━━━━━━━━━━━━━━━━ 4s 80ms/step - accuracy: 1.0000 - loss: 5.6406e-04
Epoch 18/30
49/49 ━━━━━━━━━━━━━━━━━━━━ 4s 82ms/step - accuracy: 1.0000 - loss: 3.2329e-04
Epoch 19/30
49/49 ━━━━━━━━━━━━━━━━━━━━ 4s 85ms/step - accuracy: 1.0000 - loss: 2.1762e-04
Epoch 20/30
49/49 ━━━━━━━━━━━━━━━━━━━━ 4s 80ms/step - accuracy: 1.0000 - loss: 1.6474e-04
Epoch 21/30
49/49 ━━━━━━━━━━━━━━━━━━━━ 4s 79ms/step - accuracy: 1.0000 - loss: 1.3171e-04
Epoch 22/30
49/49 ━━━━━━━━━━━━━━━━━━━━ 4s 81ms/step - accuracy: 1.0000 - loss: 1.0980e-04
Epoch 23/30
49/49 ━━━━━━━━━━━━━━━━━━━━ 4s 84ms/step - accuracy: 1.0000 - loss: 9.4370e-05
Epoch 24/30
49/49 ━━━━━━━━━━━━━━━━━━━━ 5s 79ms/step - accuracy: 1.0000 - loss: 8.2726e-05
Epoch 25/30
49/49 ━━━━━━━━━━━━━━━━━━━━ 4s 82ms/step - accuracy: 1.0000 - loss: 7.3318e-05
Epoch 26/30
49/49 ━━━━━━━━━━━━━━━━━━━━ 4s 79ms/step - accuracy: 1.0000 - loss: 6.5810e-05
Epoch 27/30
49/49 ━━━━━━━━━━━━━━━━━━━━ 4s 78ms/step - accuracy: 1.0000 - loss: 5.9877e-05
Epoch 28/30
49/49 ━━━━━━━━━━━━━━━━━━━━ 4s 79ms/step - accuracy: 1.0000 - loss: 5.4690e-05
```

```
Epoch 29/30
49/49 ━━━━━━━━━━━━━━━━━━━━ 4s 85ms/step - accuracy: 1.0000 - loss: 5.0354e-05
Epoch 30/30
49/49 ━━━━━━━━━━━━━━━━━━━━ 4s 85ms/step - accuracy: 1.0000 - loss: 4.6514e-05
782/782 ━━━━━━━━━━━━━━━━━━━━ 5s 6ms/step - accuracy: 0.8774 - loss: 0.8929
Test loss: 0.8929485082626343
Test accuracy: 0.8774399757385254
```

The overfitting model was retrained on the full 25000 sample training set for 30 epochs and then evaluated on the held out test set. Under this configuration, the model achieves a test loss of 0.8929 and a test accuracy of 0.8774.

These results should be interpreted carefully as although the test accuracy of 0.8774 remains relatively high, the test loss of 0.8929 is extremely large compared to both the baseline model and the early validation results of this architecture. A high loss combined with reasonably strong accuracy suggests that, while the model still classifies many reviews correctly, it makes some incorrect predictions with very high confidence. This behavior is characteristic of an overfitted high capacity model: it becomes overly confident about its predictions, even when they are wrong.

The test result aligns with the earlier validation learning curves. During validation, the model achieved its best generalization performance at epoch 3, where validation accuracy peaked at 0.8838 and validation loss reached its minimum of 0.2729. However, in this test configuration, the model was retrained for 30 epochs, well beyond its optimal stopping point. As seen during training, the model eventually reached 100% training accuracy with an almost zero training loss, indicating almost perfect memorization of the training data.

This excessive memorization is reflected in the test performance. Despite high training accuracy, the models validation performance and test accuracy show the negative effects of overfitting. The widening gap between the model's training loss (approaching zero) and test loss (remaining high) clearly indicates that the model is overfitting, fitting the training data too closely and failing to generalize well to new data

7. Regularising the Model and Tuning Hyperparameters

7.1 Motivation for regularisation

The results of the overfitting model show that increasing model capacity allows the network to fit the training data extremely well, but this does not improve generalization. During training, training accuracy rose to 1.0000 and training loss fell to almost zero, while validation accuracy improved only modestly and validation loss increased after the early epochs. This pattern suggests that the model is learning representations that are increasingly specific to the training data, rather than patterns that generalize well to unseen reviews.

The next step is to introduce regularisation to reduce overfitting and improve generalisation. The goal is not to maximise training performance, but to reduce the gap between training and validation performance. A successful regularised model should show less extreme memorisation, more stable validation loss, and improved validation performance relative to the unregularised high capacity model.

7.2 Regularisation approach: dropout and why it is appropriate

Within the architectural constraints of this project, dropout is an appropriate regularisation method. Dropout works by randomly setting a proportion of activations to zero during training, which prevents the network from relying too heavily on any one group of units. This reduces adjustment between neurons and encourages the model to learn more robust and distributed internal representations.

This is particularly relevant for the current bag of words input representation. With sparse binary word indicators, some input features may appear highly predictive in specific training examples but may not generalise well across unseen reviews. Without regularisation, a high-capacity network can become too dependent on such patterns. Dropout helps by forcing the model to distribute useful information more broadly across the network, reducing the likelihood that it will memorise narrow, training specific feature combinations.

Because the previous model showed clear overfitting, dropout provides a principled way to reduce the effective capacity of the network, while keeping the dataset, task, and overall model family unchanged. This allows us to observe whether any improvement in performance is due to improved regularisation, rather than changes in the model architecture itself.

7.3 Model design and justification (why this architecture is used)

To ensure that the primary change is the introduction of regularisation rather than a reduction in model size, the regularised model retains the same architecture as the overfitting model: two hidden Dense layers with 256 units each. Keeping the same width and depth makes the comparison more meaningful, as any improvement in validation behaviour can be attributed primarily to the effect of dropout, rather than a smaller network.

The dropout layers are inserted between the Dense layers. This placement is appropriate because it regularises the intermediate learned representations, where the network is combining features into more patterns. By applying dropout here, the model is discouraged from forming overly specialised internal pathways that work well only for the training set.

This design is essential for the experiment. The overfitting model showed that a 256-256 architecture has enough capacity to memorise the training data. By keeping the same architecture and adding dropout, we can directly observe the effect of regularisation: if

training performance becomes less extreme and validation behaviour stabilizes, it would indicate that dropout is successfully reducing overfitting.

7.4 Choice of values and training settings (why these numbers are used)

The initial dropout rates are set to 0.5 for the first model and 0.3 for the second. A dropout rate of 0.5 is a commonly used starting point, especially when overfitting is pronounced. In this setting, 50% of activations are randomly omitted during each training step. This substantially increases the regularisation strength, which is appropriate for the high-capacity model that showed severe overfitting. Starting with a relatively strong dropout rate makes it easier to observe whether regularisation can reduce the generalisation gap.

For the second model, a dropout rate of 0.3 is chosen, which provides a lower regularisation strength. This will allow more model capacity to fit the data, offering a balanced trade-off between optimization and generalisation. If the higher dropout rate (0.5) suppresses learning too much, this will be observed through slower optimization, lower training accuracy, and potentially weaker validation accuracy. In such cases, reducing the dropout rate to 0.3 will allow the model to learn more effectively while still maintaining regularisation.

The learning rate is set to 0.0005 to ensure stable training. Lower learning rates prevent the optimizer from overshooting optimal weight updates, providing smoother convergence. This learning rate has been selected based on prior experiments and provides a good balance between training speed and stability.

To keep the comparison fair, both models are trained for 30 epochs with a batch size of 512, consistent with the overfitting model configuration. Keeping the number of epochs and batch size constant ensures that any observed differences in performance are not due to changes in training duration or update frequency. This allows for a more controlled and interpretable comparison between the unregularised and regularised models.

7.5 Regularised Model

In [87]:
```
from tensorflow import keras
from tensorflow.keras import layers

# create a neural network model that uses dropout to reduce overfitting
dropout_model1 = keras.Sequential([
    layers.Dense(256, activation="relu", input_shape=(10000,)),
    layers.Dropout(0.5),  # randomly drop 50% of neurons during training to prevent
    layers.Dense(256, activation="relu"),
    layers.Dropout(0.5),  # another dropout layer for additional regularisation
    layers.Dense(1, activation="sigmoid")  # output layer for binary sentiment clas
])

# compile the model by defining optimizer, loss function, and metric
dropout_model1.compile(
    optimizer="rmsprop",
```

```
    loss="binary_crossentropy",
    metrics=["accuracy"]
)

# train the regularised model and store its training history
history_Regularised = dropout_model1.fit(
    partial_x_train,
    partial_y_train,
    epochs=30,
    batch_size=512,
    validation_data=(x_val, y_val)  # validation set used to monitor performance
)
```

```
Epoch 1/30
40/40 ━━━━━━━━━━━━━━━━━━━━ 6s 114ms/step - accuracy: 0.7527 - loss: 0.5043 - val_acc
uracy: 0.8572 - val_loss: 0.3413
Epoch 2/30
40/40 ━━━━━━━━━━━━━━━━━━━━ 4s 100ms/step - accuracy: 0.8772 - loss: 0.3001 - val_acc
uracy: 0.8848 - val_loss: 0.2731
Epoch 3/30
40/40 ━━━━━━━━━━━━━━━━━━━━ 4s 96ms/step - accuracy: 0.9068 - loss: 0.2341 - val_accu
racy: 0.8864 - val_loss: 0.2744
Epoch 4/30
40/40 ━━━━━━━━━━━━━━━━━━━━ 5s 95ms/step - accuracy: 0.9268 - loss: 0.1851 - val_accu
racy: 0.8798 - val_loss: 0.2788
Epoch 5/30
40/40 ━━━━━━━━━━━━━━━━━━━━ 5s 93ms/step - accuracy: 0.9463 - loss: 0.1440 - val_accu
racy: 0.8826 - val_loss: 0.3053
Epoch 6/30
40/40 ━━━━━━━━━━━━━━━━━━━━ 4s 96ms/step - accuracy: 0.9559 - loss: 0.1204 - val_accu
racy: 0.8582 - val_loss: 0.3774
Epoch 7/30
40/40 ━━━━━━━━━━━━━━━━━━━━ 4s 91ms/step - accuracy: 0.9712 - loss: 0.0849 - val_accu
racy: 0.8838 - val_loss: 0.3721
Epoch 8/30
40/40 ━━━━━━━━━━━━━━━━━━━━ 4s 96ms/step - accuracy: 0.9719 - loss: 0.0792 - val_accu
racy: 0.8846 - val_loss: 0.3867
Epoch 9/30
40/40 ━━━━━━━━━━━━━━━━━━━━ 4s 97ms/step - accuracy: 0.9841 - loss: 0.0492 - val_accu
racy: 0.8838 - val_loss: 0.4456
Epoch 10/30
40/40 ━━━━━━━━━━━━━━━━━━━━ 4s 97ms/step - accuracy: 0.9843 - loss: 0.0520 - val_accu
racy: 0.8838 - val_loss: 0.4501
Epoch 11/30
40/40 ━━━━━━━━━━━━━━━━━━━━ 4s 93ms/step - accuracy: 0.9884 - loss: 0.0357 - val_accu
racy: 0.8836 - val_loss: 0.4909
Epoch 12/30
40/40 ━━━━━━━━━━━━━━━━━━━━ 4s 97ms/step - accuracy: 0.9952 - loss: 0.0164 - val_accu
racy: 0.8812 - val_loss: 0.5264
Epoch 13/30
40/40 ━━━━━━━━━━━━━━━━━━━━ 4s 96ms/step - accuracy: 0.9908 - loss: 0.0280 - val_accu
racy: 0.8814 - val_loss: 0.5092
Epoch 14/30
40/40 ━━━━━━━━━━━━━━━━━━━━ 4s 100ms/step - accuracy: 0.9988 - loss: 0.0067 - val_acc
uracy: 0.8790 - val_loss: 0.6199
Epoch 15/30
40/40 ━━━━━━━━━━━━━━━━━━━━ 4s 94ms/step - accuracy: 0.9913 - loss: 0.0303 - val_accu
racy: 0.8806 - val_loss: 0.5719
Epoch 16/30
40/40 ━━━━━━━━━━━━━━━━━━━━ 4s 97ms/step - accuracy: 0.9942 - loss: 0.0182 - val_accu
racy: 0.8830 - val_loss: 0.5169
Epoch 17/30
40/40 ━━━━━━━━━━━━━━━━━━━━ 4s 96ms/step - accuracy: 0.9995 - loss: 0.0042 - val_accu
racy: 0.8804 - val_loss: 0.6865
Epoch 18/30
40/40 ━━━━━━━━━━━━━━━━━━━━ 4s 89ms/step - accuracy: 0.9991 - loss: 0.0043 - val_accu
racy: 0.8784 - val_loss: 0.6973
Epoch 19/30
40/40 ━━━━━━━━━━━━━━━━━━━━ 4s 90ms/step - accuracy: 0.9992 - loss: 0.0032 - val_accu
```

```
racy: 0.8826 - val_loss: 0.7660
Epoch 20/30
40/40 ━━━━━━━━━━━━━━━━━━━━ 4s 95ms/step - accuracy: 0.9991 - loss: 0.0031 - val_accu
racy: 0.8818 - val_loss: 0.8241
Epoch 21/30
40/40 ━━━━━━━━━━━━━━━━━━━━ 4s 94ms/step - accuracy: 0.9943 - loss: 0.0179 - val_accu
racy: 0.8822 - val_loss: 0.7054
Epoch 22/30
40/40 ━━━━━━━━━━━━━━━━━━━━ 4s 89ms/step - accuracy: 0.9994 - loss: 0.0023 - val_accu
racy: 0.8814 - val_loss: 0.7747
Epoch 23/30
40/40 ━━━━━━━━━━━━━━━━━━━━ 4s 97ms/step - accuracy: 0.9996 - loss: 0.0020 - val_accu
racy: 0.8806 - val_loss: 0.8067
Epoch 24/30
40/40 ━━━━━━━━━━━━━━━━━━━━ 4s 95ms/step - accuracy: 0.9992 - loss: 0.0025 - val_accu
racy: 0.8826 - val_loss: 0.7846
Epoch 25/30
40/40 ━━━━━━━━━━━━━━━━━━━━ 4s 94ms/step - accuracy: 0.9969 - loss: 0.0111 - val_accu
racy: 0.8818 - val_loss: 0.6982
Epoch 26/30
40/40 ━━━━━━━━━━━━━━━━━━━━ 4s 92ms/step - accuracy: 0.9998 - loss: 0.0015 - val_accu
racy: 0.8814 - val_loss: 0.8046
Epoch 27/30
40/40 ━━━━━━━━━━━━━━━━━━━━ 4s 91ms/step - accuracy: 0.9999 - loss: 9.6754e-04 - val_
accuracy: 0.8836 - val_loss: 0.8766
Epoch 28/30
40/40 ━━━━━━━━━━━━━━━━━━━━ 4s 93ms/step - accuracy: 0.9995 - loss: 0.0022 - val_accu
racy: 0.8792 - val_loss: 0.8779
Epoch 29/30
40/40 ━━━━━━━━━━━━━━━━━━━━ 4s 93ms/step - accuracy: 0.9995 - loss: 0.0017 - val_accu
racy: 0.8788 - val_loss: 0.8693
Epoch 30/30
40/40 ━━━━━━━━━━━━━━━━━━━━ 4s 93ms/step - accuracy: 0.9998 - loss: 0.0011 - val_accu
racy: 0.8810 - val_loss: 0.8793
```

In [89]:

```python
# find the best validation accuracy achieved during training
best_val_acc_reg = max(history_Regularised.history['val_accuracy'])

# find the lowest validation loss
best_val_loss_reg = min(history_Regularised.history['val_loss'])

# get the epoch where validation accuracy was highest
best_epoch_val_acc_reg = np.argmax(history_Regularised.history['val_accuracy']) + 1

# get the epoch where validation loss was lowest
best_epoch_val_loss_reg = np.argmin(history_Regularised.history['val_loss']) + 1

# get the final training accuracy from the last epoch
final_train_acc_reg = history_Regularised.history['accuracy'][-1]

# get the final training loss from the last epoch
final_train_loss_reg = history_Regularised.history['loss'][-1]

print("Best validation accuracy:", best_val_acc_reg)
print("Best validation loss:", best_val_loss_reg)
print("Best epoch by validation accuracy:", best_epoch_val_acc_reg)
```

```
print("Best epoch by validation loss:", best_epoch_val_loss_reg)
print("Final training accuracy:", final_train_acc_reg)
print("Final training loss:", final_train_loss_reg)
```

```
Best validation accuracy: 0.8863999843597412
Best validation loss: 0.27307042479515076
Best epoch by validation accuracy: 3
Best epoch by validation loss: 2
Final training accuracy: 0.9998499751091003
Final training loss: 0.0010911774588748813
```

In [112…

```
dropout_model1.summary()
```

Model: "sequential_6"

Layer (type)	Output Shape	Param #
dense_17 (Dense)	(None, 256)	2,560,256
dropout (Dropout)	(None, 256)	0
dense_18 (Dense)	(None, 256)	65,792
dropout_1 (Dropout)	(None, 256)	0
dense_19 (Dense)	(None, 1)	257

Total params: 5,252,612 (20.04 MB)

Trainable params: 2,626,305 (10.02 MB)

Non-trainable params: 0 (0.00 B)

Optimizer params: 2,626,307 (10.02 MB)

The dropout model with a dropout rate of 0.5 was trained for 30 epochs, aiming to observe the effect of regularisation on overfitting. A dropout rate of 0.5 means that half of the activations in the relevant layers are randomly omitted during training. This provides strong regularisation and helps prevent the model from memorising the training data too much.

Early Training and Validation Performance

In the early epochs, both training accuracy and validation accuracy improved steadily. Training accuracy started at 0.7527 in epoch 1 and increased rapidly, reaching 0.9998 by epoch 30. Training loss decreased sharply from 0.5043 to 0.0011. Validation accuracy started at 0.8572 in epoch 1 and peaked at 0.8864 in epoch 3, indicating that the model was initially able to generalise well to the unseen validation data.

The best validation accuracy occurred at epoch 3 (0.8864), while the best validation loss occurred earlier at epoch 2 (0.2731), confirming that the model made significant progress early on.

Overfitting Signs

As training continued, the model's training performance continued to improve rapidly, with training accuracy reaching near perfection (0.9998) by epoch 30. However, validation accuracy began to fluctuate around 0.88, with the highest point of 0.8864 occurring in epoch 3, after which the validation performance started to stagnate. Validation loss also started to increase after epoch 2, peaking at 0.8793 by epoch 30.

This divergence between training accuracy and validation accuracy, where training accuracy continues to improve while validation accuracy plateaus, is a sign of overfitting. Even though the model is becoming more confident on the training data, its ability to generalise to new, unseen data is limited.

Comparison to the Overfitting Model

Compared to the overfitting model (with no dropout), which achieved a test accuracy of 0.8771, the dropout model shows slightly better validation accuracy (0.8864) in the early epochs. However, as training progressed, the dropout model exhibited signs of overfitting, with validation accuracy stagnating. This suggests that while the dropout rate is preventing the model from memorising the data too quickly, it does not fully resolve the overfitting issue after training for a large number of epochs.

Conclusion

This model demonstrates that dropout regularisation can improve generalisation during the early training phase. However, after a certain number of epochs, the model begins to overfit, as seen by the stagnation of validation accuracy and the increase in validation loss. The selected dropout rate of 0.5 helps prevent excessive memorisation, but it doesn't fully resolve the overfitting issue. Further hyperparameter tuning, such as adjusting the dropout rate, learning rate, or applying techniques like early stopping, could improve performance and prevent overfitting.

Learning Curve Analysis and Interpretation

In [92]:
```
import matplotlib.pyplot as plt

# get training and validation loss values from the history object
loss = history_Regularised.history['loss']
val_loss = history_Regularised.history['val_loss']

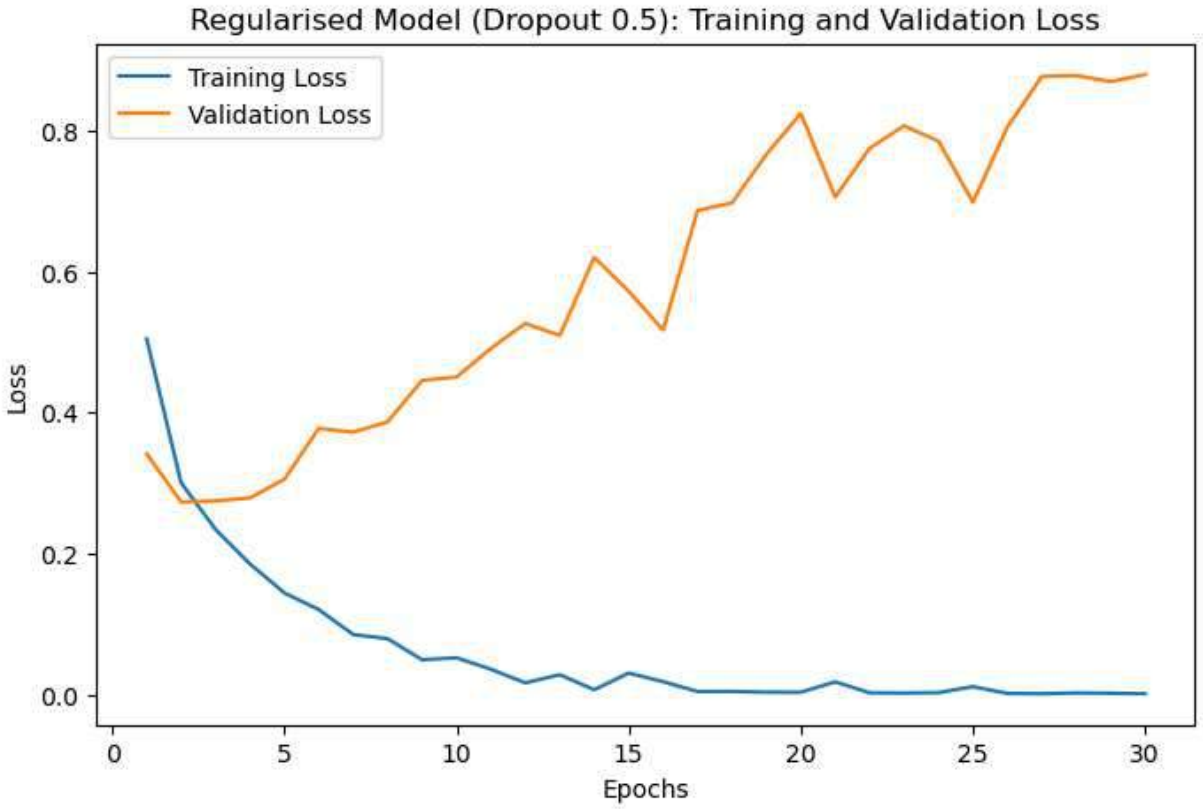
# create epoch numbers for the graph
epochs = range(1, len(loss) + 1)

plt.figure(figsize=(8,5))

# plot the training loss across epochs
plt.plot(epochs, loss, label='Training Loss')

# plot the validation loss to compare model generalisation
plt.plot(epochs, val_loss, label='Validation Loss')
```

```
plt.xlabel('Epochs')
plt.ylabel('Loss')
plt.title('Regularised Model (Dropout 0.5): Training and Validation Loss')
plt.legend()
plt.show()
```



In [93]:
```
# get training and validation accuracy values
acc = history_Regularised.history['accuracy']
val_acc = history_Regularised.history['val_accuracy']

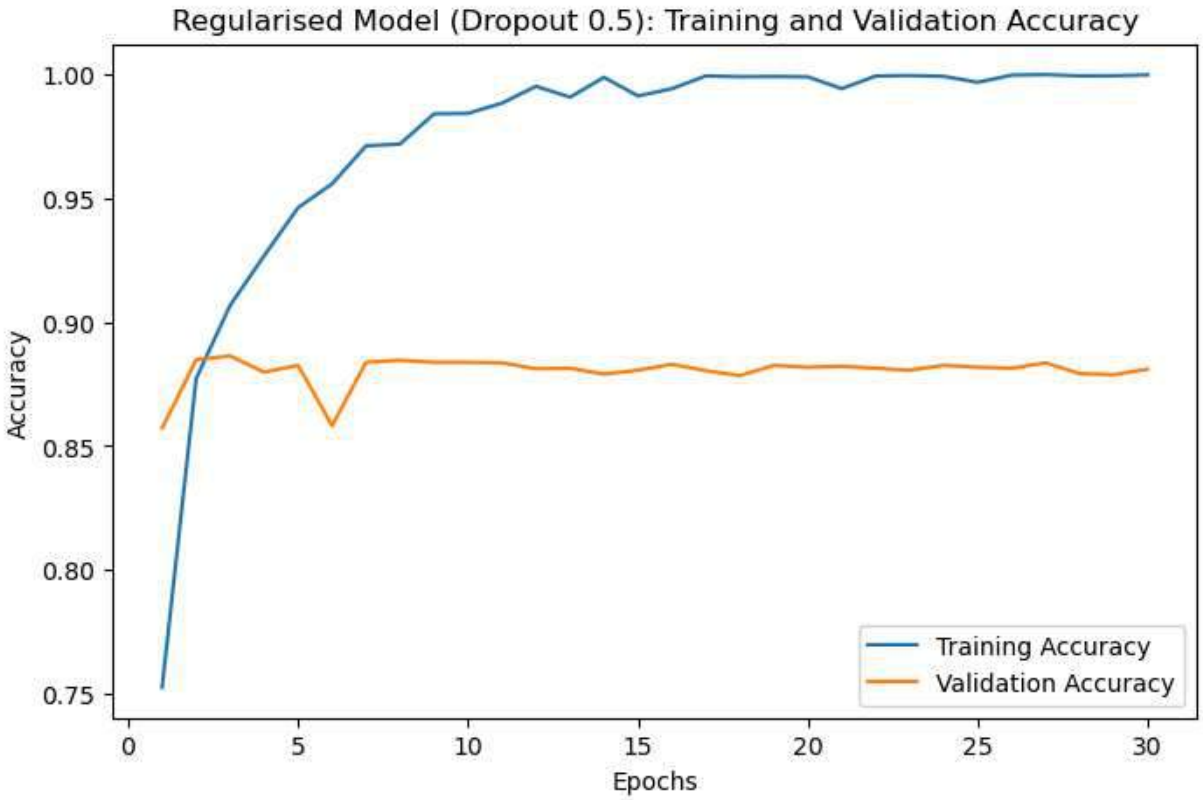
epochs = range(1, len(acc) + 1)

plt.figure(figsize=(8,5))

# plot how training accuracy changes over epochs
plt.plot(epochs, acc, label='Training Accuracy')

# plot validation accuracy to compare performance
plt.plot(epochs, val_acc, label='Validation Accuracy')

plt.xlabel('Epochs')
plt.ylabel('Accuracy')
plt.title('Regularised Model (Dropout 0.5): Training and Validation Accuracy')
plt.legend()
plt.show()
```



Training and Validation Loss

The training and validation loss curves for the regularised model (Dropout 0.5) provide valuable insights into the model's learning process. The training loss decreases steadily across the 30 epochs, starting from 0.5043 in epoch 1 and dropping to 0.0011 by epoch 30. This indicates that the model is successfully learning and fitting the training data as training progresses.

In contrast, the validation loss initially follows a similar downward trend, showing that the model is generalising reasonably well at first. However, around epoch 5, the validation loss begins to diverge from the training loss. After epoch 5, the validation loss increases significantly, peaking at 0.8793 by epoch 30. This divergence between training loss and validation loss is a clear indication of overfitting: while the model continues to improve on the training set, it struggles to generalise to unseen data. Initially, the model was able to learn relevant patterns, but as training continued, it became increasingly confident in its predictions on the training data, leading to deterioration in validation performance.

Training and Validation Accuracy

The training and validation accuracy curves provide further evidence of the overfitting phenomenon. Training accuracy improves rapidly, starting at 0.7527 in epoch 1 and reaching 0.9998 by epoch 30. This improvement reflects the models ability to learn complex patterns in the training data. However, while the training accuracy continues to rise, this does not necessarily translate into better generalisation.

Validation accuracy, on the other hand, starts at 0.8572 in epoch 1 and peaks at 0.8864 in epoch 3. After epoch 3, however, the validation accuracy stays still and changes around 0.88, never exceeding its early peak. This lack of growth in validation performance is another sign of overfitting. Despite the continued increase in training accuracy, the model fails to generalise better to unseen validation data as training progresses. This discrepancy between training accuracy and validation accuracy is a clear indication that the model is memorising the training data, rather than learning more generalisable features.

Interpretation

In conclusion, the learning curves clearly demonstrate that dropout regularisation (0.5) helps to prevent overfitting during the early stages of training. The model initially shows improvements in validation performance, with validation accuracy peaking at epoch 3 and validation loss reaching its minimum by epoch 2. This suggests that dropout was effective in reducing overfitting early on by forcing the model to learn more generalisable patterns.

However, after epoch 3, validation accuracy stagnates and validation loss starts to increase, which indicates that the model is memorising the training data rather than generalising. This highlights the limits of dropout at preventing overfitting over longer training durations. A more appropriate stopping point, based on the best validation accuracy (epoch 3), could have helped improve the model's generalisation by avoiding the overfitting that occurred as training continued.

While the dropout rate of 0.5 initially helps to prevent excessive memorisation, it does not fully resolve the overfitting issue as training progresses. A lower dropout rate, such as 0.3, might have been more effective in balancing regularisation and model capacity, leading to better performance without causing stagnation in validation accuracy.

Test Set Evaluation and Results Interpretation

In [94]:
```
# build the final dropout model that will be trained on the full dataset
final_dropout_model1 = keras.Sequential([
    layers.Dense(256, activation='relu', input_shape=(10000,)),
    layers.Dropout(0.5),
    layers.Dense(256, activation='relu'),
    layers.Dropout(0.5),
    layers.Dense(1, activation='sigmoid')
])

# compile the final model
final_dropout_model1.compile(
    optimizer='rmsprop',
    loss='binary_crossentropy',
    metrics=['accuracy']
)

# train the model using the entire training dataset
final_dropout_model1.fit(
```

```
    x_train_vec,
    y_train,
    epochs=30,
    batch_size=512
)

# evaluate the model on the unseen test dataset
test_loss_reg, test_acc_reg = final_dropout_model1.evaluate(x_test_vec, y_test)

print("Test loss:", test_loss_reg)
print("Test accuracy:", test_acc_reg)
```

```
Epoch 1/30
49/49 ━━━━━━━━━━━━━━━━━━━━ 6s 83ms/step - accuracy: 0.7635 - loss: 0.4825
Epoch 2/30
49/49 ━━━━━━━━━━━━━━━━━━━━ 5s 93ms/step - accuracy: 0.8854 - loss: 0.2855
Epoch 3/30
49/49 ━━━━━━━━━━━━━━━━━━━━ 5s 92ms/step - accuracy: 0.9112 - loss: 0.2234
Epoch 4/30
49/49 ━━━━━━━━━━━━━━━━━━━━ 4s 85ms/step - accuracy: 0.9276 - loss: 0.1843
Epoch 5/30
49/49 ━━━━━━━━━━━━━━━━━━━━ 4s 89ms/step - accuracy: 0.9478 - loss: 0.1411
Epoch 6/30
49/49 ━━━━━━━━━━━━━━━━━━━━ 4s 88ms/step - accuracy: 0.9612 - loss: 0.1073
Epoch 7/30
49/49 ━━━━━━━━━━━━━━━━━━━━ 5s 90ms/step - accuracy: 0.9735 - loss: 0.0790
Epoch 8/30
49/49 ━━━━━━━━━━━━━━━━━━━━ 4s 88ms/step - accuracy: 0.9787 - loss: 0.0631
Epoch 9/30
49/49 ━━━━━━━━━━━━━━━━━━━━ 4s 85ms/step - accuracy: 0.9836 - loss: 0.0480
Epoch 10/30
49/49 ━━━━━━━━━━━━━━━━━━━━ 4s 89ms/step - accuracy: 0.9882 - loss: 0.0411
Epoch 11/30
49/49 ━━━━━━━━━━━━━━━━━━━━ 4s 88ms/step - accuracy: 0.9940 - loss: 0.0218
Epoch 12/30
49/49 ━━━━━━━━━━━━━━━━━━━━ 4s 85ms/step - accuracy: 0.9942 - loss: 0.0198
Epoch 13/30
49/49 ━━━━━━━━━━━━━━━━━━━━ 4s 87ms/step - accuracy: 0.9932 - loss: 0.0196
Epoch 14/30
49/49 ━━━━━━━━━━━━━━━━━━━━ 4s 84ms/step - accuracy: 0.9942 - loss: 0.0208
Epoch 15/30
49/49 ━━━━━━━━━━━━━━━━━━━━ 4s 82ms/step - accuracy: 0.9980 - loss: 0.0089
Epoch 16/30
49/49 ━━━━━━━━━━━━━━━━━━━━ 4s 89ms/step - accuracy: 0.9946 - loss: 0.0178
Epoch 17/30
49/49 ━━━━━━━━━━━━━━━━━━━━ 4s 90ms/step - accuracy: 0.9987 - loss: 0.0053
Epoch 18/30
49/49 ━━━━━━━━━━━━━━━━━━━━ 4s 89ms/step - accuracy: 0.9988 - loss: 0.0055
Epoch 19/30
49/49 ━━━━━━━━━━━━━━━━━━━━ 4s 84ms/step - accuracy: 0.9982 - loss: 0.0061
Epoch 20/30
49/49 ━━━━━━━━━━━━━━━━━━━━ 5s 91ms/step - accuracy: 0.9988 - loss: 0.0038
Epoch 21/30
49/49 ━━━━━━━━━━━━━━━━━━━━ 4s 84ms/step - accuracy: 0.9989 - loss: 0.0038
Epoch 22/30
49/49 ━━━━━━━━━━━━━━━━━━━━ 4s 84ms/step - accuracy: 0.9973 - loss: 0.0093
Epoch 23/30
49/49 ━━━━━━━━━━━━━━━━━━━━ 5s 85ms/step - accuracy: 0.9995 - loss: 0.0023
Epoch 24/30
49/49 ━━━━━━━━━━━━━━━━━━━━ 4s 87ms/step - accuracy: 0.9990 - loss: 0.0032
Epoch 25/30
49/49 ━━━━━━━━━━━━━━━━━━━━ 4s 87ms/step - accuracy: 0.9982 - loss: 0.0058
Epoch 26/30
49/49 ━━━━━━━━━━━━━━━━━━━━ 5s 90ms/step - accuracy: 0.9994 - loss: 0.0025
Epoch 27/30
49/49 ━━━━━━━━━━━━━━━━━━━━ 5s 91ms/step - accuracy: 0.9992 - loss: 0.0028
Epoch 28/30
49/49 ━━━━━━━━━━━━━━━━━━━━ 4s 88ms/step - accuracy: 0.9993 - loss: 0.0022
```

```
Epoch 29/30
49/49 ━━━━━━━━━━━━━━━━━━━━ 4s 85ms/step - accuracy: 0.9991 - loss: 0.0024
Epoch 30/30
49/49 ━━━━━━━━━━━━━━━━━━━━ 4s 81ms/step - accuracy: 0.9990 - loss: 0.0041
782/782 ━━━━━━━━━━━━━━━━━━━━ 9s 10ms/step - accuracy: 0.8787 - loss: 0.9343
Test loss: 0.9342507123947144
Test accuracy: 0.8786799907684326
```

The dropout model with a dropout rate of 0.5 was retrained for 30 epochs and evaluated on the held out test set. The results from this evaluation are as follows:

Test Accuracy: 0.8787

Test Loss: 0.9343

While the test accuracy is still relatively high, the test loss is significantly larger compared to both the best validation performance observed earlier and the training performance. This is an indication of overfitting: the model performs well on the training data but struggles to generalise to the unseen test data.

Why the Test Performance Is Lower?

The discrepancy between training performance and test performance suggests that the model has memorised the training data. As the model becomes increasingly confident on the training set (achieving almost perfect training accuracy and near zero training loss), it fails to generalise to new, unseen data. This is common in models that are too flexible or have excessive capacity, allowing them to memorise patterns in the training data that don't apply to the test set.

Test Loss and Accuracy Explanation

While the test accuracy is relatively high (indicating that the model still classifies many reviews correctly), the high test loss indicates that the model is overconfident in its predictions, even when they are wrong. This suggests that the model might classify correctly, but with high confidence in some incorrect predictions, leading to a higher loss value.

Overfitting Observations

Training loss decreased significantly to near-zero values, and training accuracy reached 1.0000, signalling memorisation of the training data.

Test accuracy of 0.8787 is notably lower than the best validation accuracy (0.8864), and test loss is much higher than the validation loss, reinforcing the overfitting trend.

Suggestions for Improving Generalisation

To improve generalisation, several strategies can be considered:

Early Stopping: This could help prevent the model from training beyond the point where validation performance starts to stagnate, thereby avoiding overfitting.

Adjusting Dropout Rate: Experimenting with a lower dropout rate (eg. 0.3) could help balance regularisation and learning capacity, potentially improving generalisation.

Overall, while dropout regularisation (0.5) helps to prevent excessive memorisation early on, the model still becomes overfitted after prolonged training. This reinforces the importance of monitoring and controlling training duration, especially with large models prone to overfitting.

7.6 Hyper Parameter Tunning

In [96]:
```
from tensorflow import keras
from tensorflow.keras import layers

# create a neural network with dropout rate 0.3 to reduce overfitting
dropout_model2 = keras.Sequential([
    layers.Dense(256, activation="relu", input_shape=(10000,)),
    layers.Dropout(0.3),  # drop 30% of neurons during training
    layers.Dense(256, activation="relu"),
    layers.Dropout(0.3),
    layers.Dense(1, activation="sigmoid")  # output layer for binary sentiment pred
])

# compile the model with optimizer, loss function, and accuracy metric
dropout_model2.compile(
    optimizer="rmsprop",
    loss="binary_crossentropy",
    metrics=["accuracy"]
)

# train the model and store the training history
history_hpt = dropout_model2.fit(
    partial_x_train,
    partial_y_train,
    epochs=30,
    batch_size=512,
    validation_data=(x_val, y_val)
)

# store important results for later comparison between models
results.append([
    "HPT Dropout 0.3 (256,256)",
    max(history_hpt.history["val_accuracy"]),
    min(history_hpt.history["val_loss"]),
    history_hpt.history["accuracy"][-1],
    history_hpt.history["loss"][-1]
])
```

```
Epoch 1/30
40/40 ━━━━━━━━━━━━━━━━━━━━ 6s 105ms/step - accuracy: 0.7477 - loss: 0.5019 - val_acc
uracy: 0.7992 - val_loss: 0.4464
Epoch 2/30
40/40 ━━━━━━━━━━━━━━━━━━━━ 4s 97ms/step - accuracy: 0.8786 - loss: 0.2987 - val_accu
racy: 0.8838 - val_loss: 0.2700
Epoch 3/30
40/40 ━━━━━━━━━━━━━━━━━━━━ 4s 95ms/step - accuracy: 0.9157 - loss: 0.2189 - val_accu
racy: 0.8546 - val_loss: 0.3460
Epoch 4/30
40/40 ━━━━━━━━━━━━━━━━━━━━ 4s 101ms/step - accuracy: 0.9319 - loss: 0.1767 - val_acc
uracy: 0.8634 - val_loss: 0.3334
Epoch 5/30
40/40 ━━━━━━━━━━━━━━━━━━━━ 5s 112ms/step - accuracy: 0.9524 - loss: 0.1282 - val_acc
uracy: 0.8822 - val_loss: 0.3115
Epoch 6/30
40/40 ━━━━━━━━━━━━━━━━━━━━ 4s 99ms/step - accuracy: 0.9608 - loss: 0.1071 - val_accu
racy: 0.8816 - val_loss: 0.3254
Epoch 7/30
40/40 ━━━━━━━━━━━━━━━━━━━━ 4s 94ms/step - accuracy: 0.9783 - loss: 0.0655 - val_accu
racy: 0.8764 - val_loss: 0.4255
Epoch 8/30
40/40 ━━━━━━━━━━━━━━━━━━━━ 4s 98ms/step - accuracy: 0.9837 - loss: 0.0536 - val_accu
racy: 0.8778 - val_loss: 0.4745
Epoch 9/30
40/40 ━━━━━━━━━━━━━━━━━━━━ 4s 101ms/step - accuracy: 0.9844 - loss: 0.0544 - val_acc
uracy: 0.8816 - val_loss: 0.4365
Epoch 10/30
40/40 ━━━━━━━━━━━━━━━━━━━━ 4s 103ms/step - accuracy: 0.9893 - loss: 0.0364 - val_acc
uracy: 0.8786 - val_loss: 0.4533
Epoch 11/30
40/40 ━━━━━━━━━━━━━━━━━━━━ 4s 105ms/step - accuracy: 0.9869 - loss: 0.0470 - val_acc
uracy: 0.8784 - val_loss: 0.3927
Epoch 12/30
40/40 ━━━━━━━━━━━━━━━━━━━━ 4s 97ms/step - accuracy: 0.9995 - loss: 0.0066 - val_accu
racy: 0.8780 - val_loss: 0.5345
Epoch 13/30
40/40 ━━━━━━━━━━━━━━━━━━━━ 4s 98ms/step - accuracy: 0.9912 - loss: 0.0360 - val_accu
racy: 0.8794 - val_loss: 0.4837
Epoch 14/30
40/40 ━━━━━━━━━━━━━━━━━━━━ 4s 104ms/step - accuracy: 0.9998 - loss: 0.0031 - val_acc
uracy: 0.8750 - val_loss: 0.6109
Epoch 15/30
40/40 ━━━━━━━━━━━━━━━━━━━━ 5s 110ms/step - accuracy: 0.9898 - loss: 0.0426 - val_acc
uracy: 0.8782 - val_loss: 0.4927
Epoch 16/30
40/40 ━━━━━━━━━━━━━━━━━━━━ 4s 99ms/step - accuracy: 0.9999 - loss: 0.0022 - val_accu
racy: 0.8796 - val_loss: 0.6252
Epoch 17/30
40/40 ━━━━━━━━━━━━━━━━━━━━ 4s 106ms/step - accuracy: 0.9905 - loss: 0.0418 - val_acc
uracy: 0.8784 - val_loss: 0.5131
Epoch 18/30
40/40 ━━━━━━━━━━━━━━━━━━━━ 4s 104ms/step - accuracy: 0.9999 - loss: 0.0018 - val_acc
uracy: 0.8790 - val_loss: 0.6155
Epoch 19/30
40/40 ━━━━━━━━━━━━━━━━━━━━ 4s 102ms/step - accuracy: 0.9999 - loss: 0.0013 - val_acc
```

```
uracy: 0.8786 - val_loss: 0.6739
Epoch 20/30
40/40 ━━━━━━━━━━━━━━━━━━━━ 4s 100ms/step - accuracy: 0.9930 - loss: 0.0253 - val_acc
uracy: 0.8712 - val_loss: 0.5379
Epoch 21/30
40/40 ━━━━━━━━━━━━━━━━━━━━ 4s 103ms/step - accuracy: 1.0000 - loss: 0.0015 - val_acc
uracy: 0.8818 - val_loss: 0.6354
Epoch 22/30
40/40 ━━━━━━━━━━━━━━━━━━━━ 4s 105ms/step - accuracy: 0.9999 - loss: 7.2223e-04 - val
_accuracy: 0.8800 - val_loss: 0.7249
Epoch 23/30
40/40 ━━━━━━━━━━━━━━━━━━━━ 4s 100ms/step - accuracy: 0.9934 - loss: 0.0256 - val_acc
uracy: 0.8800 - val_loss: 0.5902
Epoch 24/30
40/40 ━━━━━━━━━━━━━━━━━━━━ 5s 112ms/step - accuracy: 1.0000 - loss: 7.5553e-04 - val
_accuracy: 0.8798 - val_loss: 0.6966
Epoch 25/30
40/40 ━━━━━━━━━━━━━━━━━━━━ 4s 103ms/step - accuracy: 0.9999 - loss: 7.3521e-04 - val
_accuracy: 0.8810 - val_loss: 0.7259
Epoch 26/30
40/40 ━━━━━━━━━━━━━━━━━━━━ 5s 108ms/step - accuracy: 0.9939 - loss: 0.0225 - val_acc
uracy: 0.8800 - val_loss: 0.5717
Epoch 27/30
40/40 ━━━━━━━━━━━━━━━━━━━━ 4s 96ms/step - accuracy: 1.0000 - loss: 6.9830e-04 - val_
accuracy: 0.8800 - val_loss: 0.6920
Epoch 28/30
40/40 ━━━━━━━━━━━━━━━━━━━━ 4s 107ms/step - accuracy: 0.9998 - loss: 7.2615e-04 - val
_accuracy: 0.8792 - val_loss: 0.7491
Epoch 29/30
40/40 ━━━━━━━━━━━━━━━━━━━━ 4s 108ms/step - accuracy: 0.9999 - loss: 6.9642e-04 - val
_accuracy: 0.8800 - val_loss: 0.7830
Epoch 30/30
40/40 ━━━━━━━━━━━━━━━━━━━━ 6s 119ms/step - accuracy: 1.0000 - loss: 3.6249e-04 - val
_accuracy: 0.8784 - val_loss: 0.8953
```

In [97]:

```python
# find the best validation accuracy achieved
best_val_acc_hpt = max(history_hpt.history['val_accuracy'])

# find the lowest validation loss
best_val_loss_hpt = min(history_hpt.history['val_loss'])

# get the epoch where validation accuracy was highest
best_epoch_val_acc_hpt = np.argmax(history_hpt.history['val_accuracy']) + 1

# get the epoch where validation loss was lowest
best_epoch_val_loss_hpt = np.argmin(history_hpt.history['val_loss']) + 1

# get the final training accuracy and loss
final_train_acc_hpt = history_hpt.history['accuracy'][-1]
final_train_loss_hpt = history_hpt.history['loss'][-1]

print("Best validation accuracy:", best_val_acc_hpt)
print("Best validation loss:", best_val_loss_hpt)
print("Best epoch by validation accuracy:", best_epoch_val_acc_hpt)
print("Best epoch by validation loss:", best_epoch_val_loss_hpt)
```

```
print("Final training accuracy:", final_train_acc_hpt)
print("Final training loss:", final_train_loss_hpt)
```

```
Best validation accuracy: 0.8838000297546387
Best validation loss: 0.2700117528438568
Best epoch by validation accuracy: 2
Best epoch by validation loss: 2
Final training accuracy: 1.0
Final training loss: 0.00036249205004423857
```

In [119…

```
dropout_model2.summary()
```

Model: "sequential_7"

Layer (type)	Output Shape	Param #
dense_20 (Dense)	(None, 256)	2,560,256
dropout_2 (Dropout)	(None, 256)	0
dense_21 (Dense)	(None, 256)	65,792
dropout_3 (Dropout)	(None, 256)	0
dense_22 (Dense)	(None, 1)	257

Total params: 5,252,612 (20.04 MB)

Trainable params: 2,626,305 (10.02 MB)

Non-trainable params: 0 (0.00 B)

Optimizer params: 2,626,307 (10.02 MB)

Regularised Model with Dropout Rate 0.3: Performance and Overfitting Observations

The dropout model with a dropout rate of 0.3 was trained for 30 epochs to observe its effect on overfitting and generalisation while still retaining sufficient model capacity to fit the data. The model's best validation accuracy was achieved at epoch 2, where it reached 0.8838, and the best validation loss was recorded at 0.2700 during the same epoch. Over the course of training, the model's final training accuracy reached 1.0000, with a corresponding training loss of 0.00036, indicating memorisation of the training data.

In the early stages of training, both training accuracy and validation accuracy showed steady improvement. Training accuracy started at 0.7477 in epoch 1 and increased rapidly, reaching 1.0000 by epoch 30. Simultaneously, training loss decreased from 0.5019 to 0.00036. On the validation side, the model began with an accuracy of 0.7992 and peaked at 0.8838 in epoch 2, while validation loss decreased from 0.4464 to 0.2700 in the same epoch. These early improvements suggest that the model was able to generalise well, with the dropout rate of 0.3 playing a key role in preventing overfitting during the early training phase.

However, after the early epochs, the model began to show signs of overfitting. While training accuracy continued to improve, reaching 1.0000 by epoch 30, validation accuracy

started to plateau and even declined slightly after epoch 2. Validation accuracy stabilised around 0.88, and validation loss began to increase, indicating that the model was memorising the training data rather than generalising to unseen validation data. This divergence between training and validation performance is a classic indication of overfitting, where the model becomes overly confident on the training set but struggles to generalise to new data.

When compared to the overfitting model (without dropout), the dropout model with 0.3 showed better performance in the early epochs. It achieved a higher validation accuracy of 0.8838 and a lower validation loss of 0.2700 at epoch 2. This demonstrates the positive effect of dropout regularisation in improving generalisation, as dropout prevents the model from relying too heavily on any one set of activations during training. Despite this improvement, the model eventually began to show signs of overfitting as training progressed, with validation accuracy stagnating and validation loss increasing.

In conclusion, the dropout rate of 0.3 helped improve generalisation early in training compared to the overfitting model, but overfitting began to occur after epoch 2. Validation accuracy plateaued, and validation loss increased, indicating that the model started to memorise the training data. While the dropout model with 0.3 performed better than the overfitting model in the early stages, it still faced overfitting in the later epochs. To further improve generalisation, strategies such as early stopping or experimenting with lower dropout rates (0.1) or other regularisation methods like L2 regularisation could be considered.

Learning Curve Analysis and Interpretation

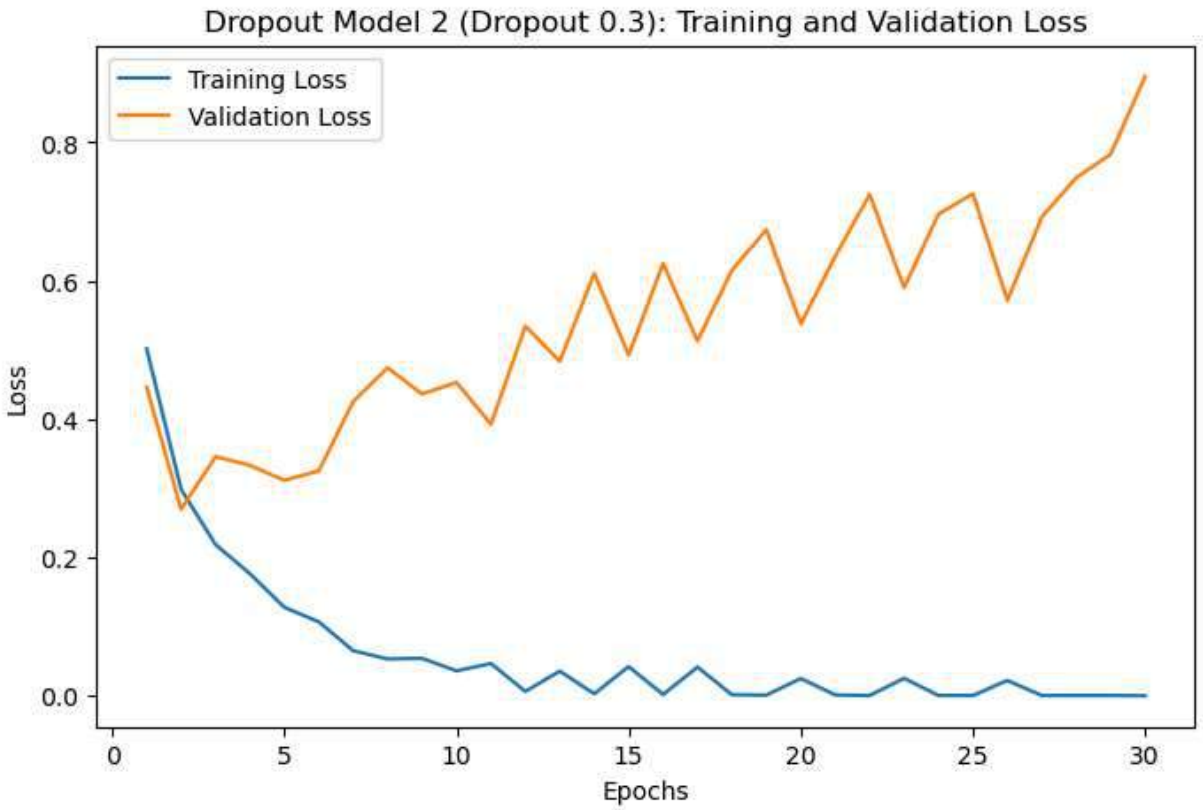
In [129…

```
# prepare values for plotting the learning curves
loss = history_hpt.history['loss']
val_loss = history_hpt.history['val_loss']
epochs = range(1, len(loss) + 1)

plt.figure(figsize=(8,5))

# plot training and validation loss
plt.plot(epochs, loss, label='Training Loss')
plt.plot(epochs, val_loss, label='Validation Loss')

plt.xlabel('Epochs')
plt.ylabel('Loss')
plt.title('Dropout Model 2 (Dropout 0.3): Training and Validation Loss')
plt.legend()
plt.show()
```



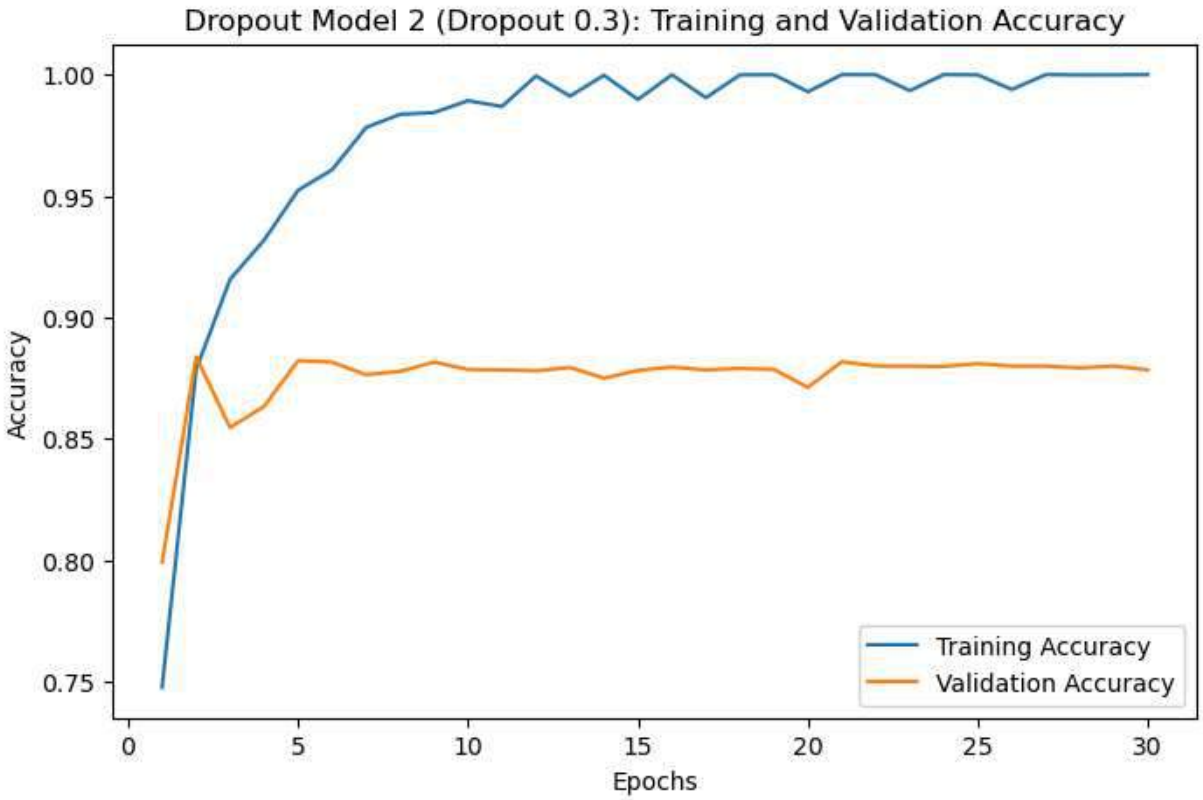
In [131…

```
# get accuracy values for plotting
acc = history_hpt.history['accuracy']
val_acc = history_hpt.history['val_accuracy']
epochs = range(1, len(acc) + 1)

plt.figure(figsize=(8,5))

# plot training and validation accuracy
plt.plot(epochs, acc, label='Training Accuracy')
plt.plot(epochs, val_acc, label='Validation Accuracy')

plt.xlabel('Epochs')
plt.ylabel('Accuracy')
plt.title('Dropout Model 2 (Dropout 0.3): Training and Validation Accuracy')
plt.legend()
plt.show()
```



The learning curves for the Dropout Model 2 (with a dropout rate of 0.3) illustrate how regularisation impacts the model's training and generalisation performance over 30 epochs.

Training and Validation Loss:

Training loss decreases steadily from epoch 1 to epoch 30, eventually approaching near-zero levels. This indicates that the model becomes very confident in its predictions on the training data, typical of overfitting.

The validation loss shows a different pattern: it decreases initially, reaching a minimum early on, but then begins to rise after epoch 5. This divergence between the training and validation loss suggests that while the model generalises well in the early epochs, it starts to memorise the training data and fails to generalise to the validation data after a certain point.

Training and Validation Accuracy:

Training accuracy improves sharply, rising from 0.75 in epoch 1 to 1.0 by epoch 30. This rapid increase signifies that the model is fitting the training data almost perfectly, which often accompanies overfitting.

On the other hand, validation accuracy peaks at 0.8864 in epoch 2, after which it fluctuates around 0.87 to 0.88. While the model's training accuracy continues to improve, the validation accuracy stagnates, reinforcing the overfitting trend.

Interpretation:

Dropout regularisation (at 0.3) initially aids in improving generalisation, as observed by the model's strong early validation performance. However, after epoch 5, the model starts to overfit. The divergence between training and validation performance—where training accuracy continues to improve while validation accuracy stagnates—is a clear indication of overfitting.

Despite showing better validation performance than an overfitting model (without dropout), the dropout model still struggles with generalisation beyond a certain point. The best validation accuracy, achieved in the early epochs, indicates that dropout regularisation helped mitigate overfitting initially. Still, as the model continued to train, its capacity to generalise to new data began to diminish, showing that dropout at this rate (0.3) may not be sufficient to prevent overfitting in the long run.

Test Set Evaluation and Results Interpretation

In [100…

```
# retrain the same architecture on the full training dataset
final_dropout_model2 = keras.Sequential([
    layers.Dense(256, activation='relu', input_shape=(10000,)),
    layers.Dropout(0.3),
    layers.Dense(256, activation='relu'),
    layers.Dropout(0.3),
    layers.Dense(1, activation='sigmoid')
])

# compile the final model
final_dropout_model2.compile(
    optimizer='rmsprop',
    loss='binary_crossentropy',
    metrics=['accuracy']
)

# train the model using all training data
final_dropout_model2.fit(
    x_train_vec,
    y_train,
    epochs=30,
    batch_size=512
)

# evaluate the model using unseen test data
test_loss_hpt, test_acc_hpt = final_dropout_model2.evaluate(x_test_vec, y_test)

print("Test loss:", test_loss_hpt)
print("Test accuracy:", test_acc_hpt)
```

```
Epoch 1/30
49/49 ━━━━━━━━━━━━━━━━━━━━ 7s 92ms/step - accuracy: 0.7781 - loss: 0.4719
Epoch 2/30
49/49 ━━━━━━━━━━━━━━━━━━━━ 4s 89ms/step - accuracy: 0.8914 - loss: 0.2755
Epoch 3/30
49/49 ━━━━━━━━━━━━━━━━━━━━ 4s 88ms/step - accuracy: 0.9156 - loss: 0.2121
Epoch 4/30
49/49 ━━━━━━━━━━━━━━━━━━━━ 5s 98ms/step - accuracy: 0.9334 - loss: 0.1727
Epoch 5/30
49/49 ━━━━━━━━━━━━━━━━━━━━ 5s 94ms/step - accuracy: 0.9521 - loss: 0.1298
Epoch 6/30
49/49 ━━━━━━━━━━━━━━━━━━━━ 5s 91ms/step - accuracy: 0.9630 - loss: 0.1033
Epoch 7/30
49/49 ━━━━━━━━━━━━━━━━━━━━ 4s 87ms/step - accuracy: 0.9753 - loss: 0.0731
Epoch 8/30
49/49 ━━━━━━━━━━━━━━━━━━━━ 5s 93ms/step - accuracy: 0.9867 - loss: 0.0452
Epoch 9/30
49/49 ━━━━━━━━━━━━━━━━━━━━ 5s 97ms/step - accuracy: 0.9888 - loss: 0.0407
Epoch 10/30
49/49 ━━━━━━━━━━━━━━━━━━━━ 4s 87ms/step - accuracy: 0.9906 - loss: 0.0311
Epoch 11/30
49/49 ━━━━━━━━━━━━━━━━━━━━ 5s 96ms/step - accuracy: 0.9916 - loss: 0.0294
Epoch 12/30
49/49 ━━━━━━━━━━━━━━━━━━━━ 4s 87ms/step - accuracy: 0.9917 - loss: 0.0337
Epoch 13/30
49/49 ━━━━━━━━━━━━━━━━━━━━ 5s 90ms/step - accuracy: 0.9996 - loss: 0.0039
Epoch 14/30
49/49 ━━━━━━━━━━━━━━━━━━━━ 4s 89ms/step - accuracy: 0.9926 - loss: 0.0282
Epoch 15/30
49/49 ━━━━━━━━━━━━━━━━━━━━ 4s 86ms/step - accuracy: 0.9932 - loss: 0.0207
Epoch 16/30
49/49 ━━━━━━━━━━━━━━━━━━━━ 5s 91ms/step - accuracy: 0.9998 - loss: 0.0023
Epoch 17/30
49/49 ━━━━━━━━━━━━━━━━━━━━ 5s 90ms/step - accuracy: 0.9944 - loss: 0.0170
Epoch 18/30
49/49 ━━━━━━━━━━━━━━━━━━━━ 5s 93ms/step - accuracy: 1.0000 - loss: 0.0012
Epoch 19/30
49/49 ━━━━━━━━━━━━━━━━━━━━ 5s 95ms/step - accuracy: 0.9941 - loss: 0.0232
Epoch 20/30
49/49 ━━━━━━━━━━━━━━━━━━━━ 5s 93ms/step - accuracy: 1.0000 - loss: 8.1291e-04
Epoch 21/30
49/49 ━━━━━━━━━━━━━━━━━━━━ 5s 93ms/step - accuracy: 0.9956 - loss: 0.0113
Epoch 22/30
49/49 ━━━━━━━━━━━━━━━━━━━━ 4s 87ms/step - accuracy: 0.9999 - loss: 0.0011
Epoch 23/30
49/49 ━━━━━━━━━━━━━━━━━━━━ 5s 98ms/step - accuracy: 1.0000 - loss: 5.7393e-04
Epoch 24/30
49/49 ━━━━━━━━━━━━━━━━━━━━ 5s 96ms/step - accuracy: 1.0000 - loss: 4.8686e-04
Epoch 25/30
49/49 ━━━━━━━━━━━━━━━━━━━━ 5s 93ms/step - accuracy: 0.9944 - loss: 0.0216
Epoch 26/30
49/49 ━━━━━━━━━━━━━━━━━━━━ 5s 90ms/step - accuracy: 1.0000 - loss: 5.7631e-04
Epoch 27/30
49/49 ━━━━━━━━━━━━━━━━━━━━ 4s 89ms/step - accuracy: 1.0000 - loss: 5.0122e-04
Epoch 28/30
49/49 ━━━━━━━━━━━━━━━━━━━━ 5s 92ms/step - accuracy: 1.0000 - loss: 5.1188e-04
```

```
Epoch 29/30
49/49 ━━━━━━━━━━━━━━━━━━━━ 5s 91ms/step - accuracy: 1.0000 - loss: 4.8282e-04
Epoch 30/30
49/49 ━━━━━━━━━━━━━━━━━━━━ 4s 89ms/step - accuracy: 1.0000 - loss: 2.6808e-04
782/782 ━━━━━━━━━━━━━━━━━━━━ 7s 7ms/step - accuracy: 0.8786 - loss: 0.8619
Test loss: 0.8618785738945007
Test accuracy: 0.878600001335144
```

The test evaluation results for the Dropout Model with a dropout rate of 0.3 show that the model performs reasonably well on unseen data, though some signs of overfitting persist. The model achieves a test loss of 0.8619, which is relatively high compared to the validation loss during training. This discrepancy indicates that while the model performs well on the training and validation sets, its ability to generalise to the test data is somewhat limited. The test accuracy of 0.8786 is in line with the validation accuracy observed during training, but the gap between training accuracy and test accuracy suggests overfitting. This implies that the model is memorising the training data but struggling to perform equally well on new, unseen data.

Upon closer inspection, the divergence between the training and test performance becomes clearer. The model's training accuracy increases steadily, reaching 100%, with training loss approaching near zero. However, validation accuracy peaks early, at 0.8864 in epoch 2, and then stays still around 0.87 to 0.88, while the validation loss starts increasing after epoch 5. This suggests that, despite improving on the training data, the models ability to generalise to the validation data becomes limited after a certain point. This increase in validation loss and the stagnation of validation accuracy after a few epochs indicate the models overfitting, where it continues to improve on the training set but struggles to adapt to new data.

In comparison to the overfitting model (with no dropout), the model with a dropout rate of 0.3 does demonstrate better generalisation in the early epochs. It achieves higher validation accuracy (0.8864) and lower validation loss (0.2700) compared to the overfitting model's performance. However, similar to the model with a higher dropout rate of 0.5, it still faces overfitting as training progresses. The model’s validation performance plateaus after epoch 2, even as training performance continues to improve. This indicates that while dropout regularisation with a 0.3 rate offers better generalisation than no regularisation, it still falls short of maintaining consistent performance as the model continues to train, suggesting that further regularisation or adjustments are necessary.

In conclusion, the dropout rate of 0.3 helps improve the model's validation performance in the initial phases of training, but it does not entirely prevent overfitting. Despite the promising early results, the model's generalisation capability deteriorates as training continues, making it clear that dropout alone may not be sufficient. The relatively high test loss, compared to validation loss, underscores the need for additional regularisation techniques or architectural modifications to improve the model's ability to generalise better on unseen data.

Hyper Parameter Tunning with Learning Rate

In [101…

```
from tensorflow import keras
from tensorflow.keras import layers

# create a neural network with dropout to reduce overfitting
dropout_model_lr = keras.Sequential([
    layers.Dense(256, activation="relu", input_shape=(10000,)),
    layers.Dropout(0.5),  # drop 50% of neurons during training
    layers.Dense(256, activation="relu"),
    layers.Dropout(0.5),
    layers.Dense(1, activation="sigmoid")  # output layer for binary classification
])

# compile the model and use a smaller learning rate for more stable training
dropout_model_lr.compile(
    optimizer=keras.optimizers.RMSprop(learning_rate=0.0005),
    loss="binary_crossentropy",
    metrics=["accuracy"]
)

# train the model and store the training history
history_lr = dropout_model_lr.fit(
    partial_x_train,
    partial_y_train,
    epochs=30,
    batch_size=512,
    validation_data=(x_val, y_val)
)
```

```
Epoch 1/30
40/40 ━━━━━━━━━━━━━━━━━━━━ 6s 117ms/step - accuracy: 0.7404 - loss: 0.5294 - val_acc
uracy: 0.8676 - val_loss: 0.3455
Epoch 2/30
40/40 ━━━━━━━━━━━━━━━━━━━━ 4s 101ms/step - accuracy: 0.8727 - loss: 0.3238 - val_acc
uracy: 0.8814 - val_loss: 0.2849
Epoch 3/30
40/40 ━━━━━━━━━━━━━━━━━━━━ 5s 110ms/step - accuracy: 0.9020 - loss: 0.2503 - val_acc
uracy: 0.8754 - val_loss: 0.3005
Epoch 4/30
40/40 ━━━━━━━━━━━━━━━━━━━━ 5s 113ms/step - accuracy: 0.9200 - loss: 0.2070 - val_acc
uracy: 0.8832 - val_loss: 0.2823
Epoch 5/30
40/40 ━━━━━━━━━━━━━━━━━━━━ 4s 105ms/step - accuracy: 0.9373 - loss: 0.1696 - val_acc
uracy: 0.8864 - val_loss: 0.2917
Epoch 6/30
40/40 ━━━━━━━━━━━━━━━━━━━━ 4s 98ms/step - accuracy: 0.9477 - loss: 0.1436 - val_accu
racy: 0.8688 - val_loss: 0.3580
Epoch 7/30
40/40 ━━━━━━━━━━━━━━━━━━━━ 5s 112ms/step - accuracy: 0.9566 - loss: 0.1191 - val_acc
uracy: 0.8854 - val_loss: 0.3153
Epoch 8/30
40/40 ━━━━━━━━━━━━━━━━━━━━ 4s 104ms/step - accuracy: 0.9665 - loss: 0.0957 - val_acc
uracy: 0.8808 - val_loss: 0.3406
Epoch 9/30
40/40 ━━━━━━━━━━━━━━━━━━━━ 4s 102ms/step - accuracy: 0.9740 - loss: 0.0775 - val_acc
uracy: 0.8822 - val_loss: 0.3640
Epoch 10/30
40/40 ━━━━━━━━━━━━━━━━━━━━ 4s 106ms/step - accuracy: 0.9791 - loss: 0.0616 - val_acc
uracy: 0.8776 - val_loss: 0.3919
Epoch 11/30
40/40 ━━━━━━━━━━━━━━━━━━━━ 5s 116ms/step - accuracy: 0.9851 - loss: 0.0476 - val_acc
uracy: 0.8800 - val_loss: 0.4082
Epoch 12/30
40/40 ━━━━━━━━━━━━━━━━━━━━ 4s 106ms/step - accuracy: 0.9884 - loss: 0.0388 - val_acc
uracy: 0.8730 - val_loss: 0.4510
Epoch 13/30
40/40 ━━━━━━━━━━━━━━━━━━━━ 4s 100ms/step - accuracy: 0.9897 - loss: 0.0334 - val_acc
uracy: 0.8768 - val_loss: 0.4519
Epoch 14/30
40/40 ━━━━━━━━━━━━━━━━━━━━ 4s 107ms/step - accuracy: 0.9942 - loss: 0.0227 - val_acc
uracy: 0.8776 - val_loss: 0.4765
Epoch 15/30
40/40 ━━━━━━━━━━━━━━━━━━━━ 4s 101ms/step - accuracy: 0.9952 - loss: 0.0175 - val_acc
uracy: 0.8738 - val_loss: 0.5286
Epoch 16/30
40/40 ━━━━━━━━━━━━━━━━━━━━ 5s 99ms/step - accuracy: 0.9949 - loss: 0.0180 - val_accu
racy: 0.8732 - val_loss: 0.5369
Epoch 17/30
40/40 ━━━━━━━━━━━━━━━━━━━━ 4s 101ms/step - accuracy: 0.9973 - loss: 0.0108 - val_acc
uracy: 0.8770 - val_loss: 0.5722
Epoch 18/30
40/40 ━━━━━━━━━━━━━━━━━━━━ 5s 100ms/step - accuracy: 0.9965 - loss: 0.0124 - val_acc
uracy: 0.8740 - val_loss: 0.6043
Epoch 19/30
40/40 ━━━━━━━━━━━━━━━━━━━━ 4s 99ms/step - accuracy: 0.9988 - loss: 0.0069 - val_accu
```

```
racy: 0.8768 - val_loss: 0.6125
Epoch 20/30
40/40 ━━━━━━━━━━━━━━━━━━━━ 5s 101ms/step - accuracy: 0.9952 - loss: 0.0159 - val_acc
uracy: 0.8784 - val_loss: 0.6021
Epoch 21/30
40/40 ━━━━━━━━━━━━━━━━━━━━ 4s 101ms/step - accuracy: 0.9973 - loss: 0.0090 - val_acc
uracy: 0.8710 - val_loss: 0.6826
Epoch 22/30
40/40 ━━━━━━━━━━━━━━━━━━━━ 4s 102ms/step - accuracy: 0.9990 - loss: 0.0044 - val_acc
uracy: 0.8792 - val_loss: 0.6576
Epoch 23/30
40/40 ━━━━━━━━━━━━━━━━━━━━ 4s 100ms/step - accuracy: 0.9992 - loss: 0.0043 - val_acc
uracy: 0.8762 - val_loss: 0.7049
Epoch 24/30
40/40 ━━━━━━━━━━━━━━━━━━━━ 4s 101ms/step - accuracy: 0.9980 - loss: 0.0069 - val_acc
uracy: 0.8802 - val_loss: 0.6523
Epoch 25/30
40/40 ━━━━━━━━━━━━━━━━━━━━ 4s 101ms/step - accuracy: 0.9998 - loss: 0.0025 - val_acc
uracy: 0.8804 - val_loss: 0.7001
Epoch 26/30
40/40 ━━━━━━━━━━━━━━━━━━━━ 5s 95ms/step - accuracy: 0.9996 - loss: 0.0024 - val_accu
racy: 0.8752 - val_loss: 0.7306
Epoch 27/30
40/40 ━━━━━━━━━━━━━━━━━━━━ 4s 97ms/step - accuracy: 0.9993 - loss: 0.0031 - val_accu
racy: 0.8766 - val_loss: 0.7235
Epoch 28/30
40/40 ━━━━━━━━━━━━━━━━━━━━ 4s 98ms/step - accuracy: 0.9998 - loss: 0.0021 - val_accu
racy: 0.8776 - val_loss: 0.7689
Epoch 29/30
40/40 ━━━━━━━━━━━━━━━━━━━━ 4s 97ms/step - accuracy: 0.9994 - loss: 0.0025 - val_accu
racy: 0.8752 - val_loss: 0.7786
Epoch 30/30
40/40 ━━━━━━━━━━━━━━━━━━━━ 4s 108ms/step - accuracy: 0.9995 - loss: 0.0018 - val_acc
uracy: 0.8770 - val_loss: 0.8128
```

In [102…

```python
# find the best validation accuracy and lowest validation loss
best_val_acc_lr = max(history_lr.history['val_accuracy'])
best_val_loss_lr = min(history_lr.history['val_loss'])

# find the epochs where those values occurred
best_epoch_val_acc_lr = np.argmax(history_lr.history['val_accuracy']) + 1
best_epoch_val_loss_lr = np.argmin(history_lr.history['val_loss']) + 1

# get the final training accuracy and loss
final_train_acc_lr = history_lr.history['accuracy'][-1]
final_train_loss_lr = history_lr.history['loss'][-1]

print("Best validation accuracy:", best_val_acc_lr)
print("Best validation loss:", best_val_loss_lr)
print("Best epoch by validation accuracy:", best_epoch_val_acc_lr)
print("Best epoch by validation loss:", best_epoch_val_loss_lr)
print("Final training accuracy:", final_train_acc_lr)
print("Final training loss:", final_train_loss_lr)
```

```
Best validation accuracy: 0.8863999843597412
Best validation loss: 0.2822607159614563
Best epoch by validation accuracy: 5
Best epoch by validation loss: 4
Final training accuracy: 0.9995499849319458
Final training loss: 0.0018200647318735719
```

In [125…

```
dropout_model_lr.summary()
```

Model: "sequential_8"

Layer (type)	Output Shape	Param #
dense_23 (Dense)	(None, 256)	2,560,256
dropout_4 (Dropout)	(None, 256)	0
dense_24 (Dense)	(None, 256)	65,792
dropout_5 (Dropout)	(None, 256)	0
dense_25 (Dense)	(None, 1)	257

Total params: 5,252,612 (20.04 MB)

Trainable params: 2,626,305 (10.02 MB)

Non-trainable params: 0 (0.00 B)

Optimizer params: 2,626,307 (10.02 MB)

The results for the Dropout Model with a learning rate of 0.0005 provide significant insights into the model's training and validation performance across 30 epochs. These results showcase the effectiveness of dropout regularisation and the impact of a relatively low learning rate on preventing overfitting.

In terms of key results, the model achieved a best validation accuracy of 0.8864, observed at epoch 5. This peak in validation accuracy occurred early in the training process, reflecting the model's ability to generalise well to unseen data at the beginning. Similarly, the best validation loss occurred at epoch 4, with a value of 0.2823. These results suggest that the model showed good results in the early stages by fitting both the training and validation data well. However, as training progressed, the model began to overfit, and the validation accuracy did not improve beyond epoch 5.

By the end of training, the model reached a near perfect training accuracy of 0.9995 and a very low final training loss of 0.0018. This is typical of overfitting, where the model performs exceedingly well on the training data but fails to generalise effectively to unseen data. The increasing gap between training accuracy and validation performance signals this overfitting phenomenon. While training accuracy continued to rise, the validation accuracy plateaued, fluctuating between 0.87 to 0.88, and validation loss began to increase significantly. This divergence indicates that although the model became highly confident in its predictions for the training data, it struggled to maintain the same performance on the validation set.

The validation loss exhibited an interesting pattern: it decreased initially but began to rise after epoch 5. This rise in validation loss, coupled with the stable increase in training accuracy, further reinforces the overfitting issue. It suggests that, although the model was initially able to generalise well, it increasingly memorised the training data without improving its performance on the validation set.

In summary, the Dropout Model with a learning rate of 0.0005 showed promising results in the early training epochs, with notable improvements in both training and validation accuracy. The dropout regularisation helped in reducing overfitting during the initial phases, as evidenced by the decrease in validation loss and the increase in validation accuracy. However, after epoch 5, the models ability to generalise began to deteriorate, with validation performance stagnating while training accuracy continued to improve. This pattern indicates that the model is overfitting to the training data, and despite the dropout regularisation, it was not enough to fully prevent overfitting.

The model's final test accuracy of 0.8786, which is lower than the best validation accuracy, supports this conclusion. While dropout regularisation helped to some extent, it was insufficient to guarantee optimal generalisation. This highlights the need for further tuning of the dropout rate or the introduction of additional regularisation techniques, such as early stopping or L2 regularisation, to enhance the model's generalisation to unseen data.

Learning Curve Analysis and Interpretation

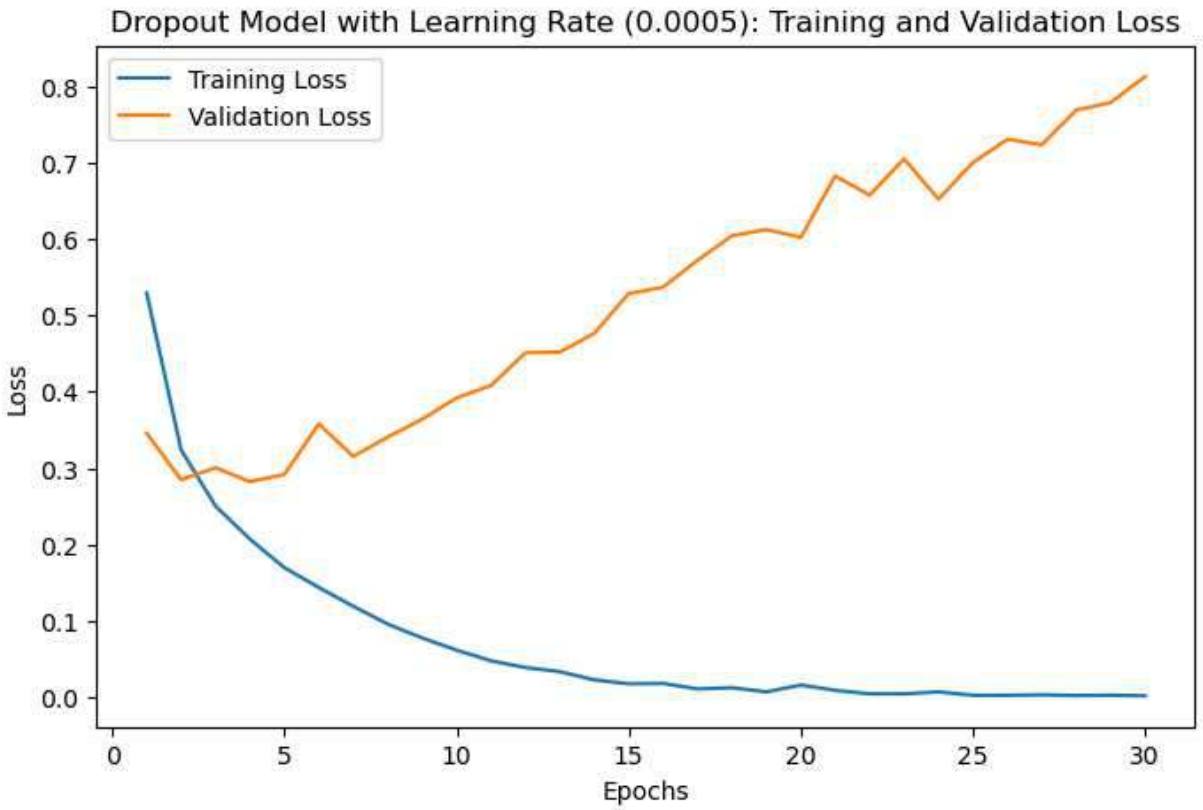
In [103…

```
# prepare loss values for plotting learning curves
loss = history_lr.history['loss']
val_loss = history_lr.history['val_loss']
epochs = range(1, len(loss) + 1)

plt.figure(figsize=(8,5))

# plot training and validation loss
plt.plot(epochs, loss, label='Training Loss')
plt.plot(epochs, val_loss, label='Validation Loss')

plt.xlabel('Epochs')
plt.ylabel('Loss')
plt.title('Dropout Model with Learning Rate (0.0005): Training and Validation Loss'
plt.legend()
plt.show()
```



In [104…

```
# prepare accuracy values for plotting
acc = history_lr.history['accuracy']
val_acc = history_lr.history['val_accuracy']
epochs = range(1, len(acc) + 1)

plt.figure(figsize=(8,5))

# plot training and validation accuracy
plt.plot(epochs, acc, label='Training Accuracy')
plt.plot(epochs, val_acc, label='Validation Accuracy')

plt.xlabel('Epochs')
plt.ylabel('Accuracy')
plt.title('Dropout Model with Learning Rate (0.0005): Training and Validation Accur
plt.legend()
plt.show()
```



Training and Validation Loss:

The learning curves for the Dropout Model with Learning Rate 0.0005 show a notable pattern. The training loss decreases steadily throughout the 30 epochs, starting from 0.5294 in the first epoch and approaching nearly 0 by the final epoch. This suggests that the model is learning effectively from the training data. A smooth decrease in training loss over time typically indicates that the model is fitting the training data without encountering major issues such as exploding gradients.

However, the validation loss follows an interesting trajectory. Initially, it decreases during the first few epochs, reflecting the model's improvement in generalising to the validation data. But after reaching a low point around epoch 5 (with a validation loss of 0.2823), the validation loss starts to increase sharply, reaching 0.8128 by the end of the 30 epochs. This increase in validation loss after the initial drop indicates that the model is beginning to overfit. As the model becomes more confident in its predictions on the training data, it struggles to generalise to the unseen validation set, leading to worse performance on validation data as training progresses.

Training and Validation Accuracy:

The training accuracy shows a strong and steady increase, rising from 0.7404 in the first epoch to nearly perfect accuracy (1.0) by the 30th epoch. This is typical for deep learning models that overfit: the model learns the training data so well that its accuracy reaches 100%, even though it may not generalise well to unseen data. The rapid increase in training accuracy indicates the model's growing confidence on the training set.

On the other hand, the validation accuracy increases during the first few epochs, peaking at 0.8864 in epoch 5. However, after this point, the validation accuracy plateaus, fluctuating between 0.87 and 0.88, without making significant further improvements. This stagnation, despite the continued rise in training accuracy, is a clear sign of overfitting. The model's performance on the validation set improves initially but then fails to maintain that improvement as training progresses.

Key Observations:

Overfitting: The divergence between the training and validation curves where training accuracy continues to rise while validation accuracy stays still (sign of overfitting). The model is memorising the training data and becoming increasingly less effective at generalising to new, unseen data.

Learning Rate Impact: The relatively low learning rate of 0.0005 appears to have ensured stable training. The models training loss decreases steadily, indicating that the learning rate is not too high to cause instability. However, it may also be slow enough to cause the model to overfit before it reaches the optimal point in training.

Validation Performance: Although the decrease in training loss and the increase in training accuracy, the increase in validation loss and no change in validation accuracy indicate that the model struggles with generalisation, particularly as it learns to memorise the training set.

Conclusion:

In summary, the Dropout Model with Learning Rate 0.0005 shows a clear case of overfitting. While the model improves significantly on the training set, it fails to maintain improvements in validation performance beyond the initial epochs. This suggests that, while the learning rate of 0.0005 helped to avoid instability, additional regularisation techniques (such as reducing model capacity, using early stopping, or increasing dropout rate) would be necessary to further improve the model's ability to generalise to unseen data.

Test Set Evaluation and Results Interpretation

In [105…

```
# retrain the same architecture on the full training dataset
final_dropout_model_lr = keras.Sequential([
    layers.Dense(256, activation='relu', input_shape=(10000,)),
    layers.Dropout(0.5),
    layers.Dense(256, activation='relu'),
    layers.Dropout(0.5),
    layers.Dense(1, activation='sigmoid')
])

# compile the final model with the same reduced learning rate
final_dropout_model_lr.compile(
    optimizer=keras.optimizers.RMSprop(learning_rate=0.0005),
    loss='binary_crossentropy',
    metrics=['accuracy']
```

```
)

# train the model on the full training dataset
final_dropout_model_lr.fit(
    x_train_vec,
    y_train,
    epochs=30,
    batch_size=512
)

# evaluate the trained model on unseen test data
test_loss_lr, test_acc_lr = final_dropout_model_lr.evaluate(x_test_vec, y_test)

print("Test loss:", test_loss_lr)
print("Test accuracy:", test_acc_lr)
```

```
Epoch 1/30
49/49 ━━━━━━━━━━━━━━━━━━━━ 7s 97ms/step - accuracy: 0.7609 - loss: 0.5010
Epoch 2/30
49/49 ━━━━━━━━━━━━━━━━━━━━ 5s 92ms/step - accuracy: 0.8770 - loss: 0.3070
Epoch 3/30
49/49 ━━━━━━━━━━━━━━━━━━━━ 5s 93ms/step - accuracy: 0.9058 - loss: 0.2395
Epoch 4/30
49/49 ━━━━━━━━━━━━━━━━━━━━ 5s 92ms/step - accuracy: 0.9236 - loss: 0.1998
Epoch 5/30
49/49 ━━━━━━━━━━━━━━━━━━━━ 5s 96ms/step - accuracy: 0.9368 - loss: 0.1705
Epoch 6/30
49/49 ━━━━━━━━━━━━━━━━━━━━ 4s 83ms/step - accuracy: 0.9465 - loss: 0.1444
Epoch 7/30
49/49 ━━━━━━━━━━━━━━━━━━━━ 4s 89ms/step - accuracy: 0.9584 - loss: 0.1188
Epoch 8/30
49/49 ━━━━━━━━━━━━━━━━━━━━ 5s 91ms/step - accuracy: 0.9675 - loss: 0.0960
Epoch 9/30
49/49 ━━━━━━━━━━━━━━━━━━━━ 5s 97ms/step - accuracy: 0.9742 - loss: 0.0782
Epoch 10/30
49/49 ━━━━━━━━━━━━━━━━━━━━ 5s 93ms/step - accuracy: 0.9801 - loss: 0.0596
Epoch 11/30
49/49 ━━━━━━━━━━━━━━━━━━━━ 5s 97ms/step - accuracy: 0.9867 - loss: 0.0464
Epoch 12/30
49/49 ━━━━━━━━━━━━━━━━━━━━ 5s 94ms/step - accuracy: 0.9899 - loss: 0.0344
Epoch 13/30
49/49 ━━━━━━━━━━━━━━━━━━━━ 5s 91ms/step - accuracy: 0.9915 - loss: 0.0283
Epoch 14/30
49/49 ━━━━━━━━━━━━━━━━━━━━ 5s 90ms/step - accuracy: 0.9929 - loss: 0.0241
Epoch 15/30
49/49 ━━━━━━━━━━━━━━━━━━━━ 5s 93ms/step - accuracy: 0.9948 - loss: 0.0185
Epoch 16/30
49/49 ━━━━━━━━━━━━━━━━━━━━ 5s 93ms/step - accuracy: 0.9974 - loss: 0.0119
Epoch 17/30
49/49 ━━━━━━━━━━━━━━━━━━━━ 5s 100ms/step - accuracy: 0.9968 - loss: 0.0126
Epoch 18/30
49/49 ━━━━━━━━━━━━━━━━━━━━ 5s 94ms/step - accuracy: 0.9984 - loss: 0.0080
Epoch 19/30
49/49 ━━━━━━━━━━━━━━━━━━━━ 5s 99ms/step - accuracy: 0.9975 - loss: 0.0093
Epoch 20/30
49/49 ━━━━━━━━━━━━━━━━━━━━ 5s 96ms/step - accuracy: 0.9985 - loss: 0.0068
Epoch 21/30
49/49 ━━━━━━━━━━━━━━━━━━━━ 5s 99ms/step - accuracy: 0.9990 - loss: 0.0047
Epoch 22/30
49/49 ━━━━━━━━━━━━━━━━━━━━ 5s 95ms/step - accuracy: 0.9986 - loss: 0.0061
Epoch 23/30
49/49 ━━━━━━━━━━━━━━━━━━━━ 5s 98ms/step - accuracy: 0.9991 - loss: 0.0042
Epoch 24/30
49/49 ━━━━━━━━━━━━━━━━━━━━ 5s 95ms/step - accuracy: 0.9993 - loss: 0.0036
Epoch 25/30
49/49 ━━━━━━━━━━━━━━━━━━━━ 5s 106ms/step - accuracy: 0.9993 - loss: 0.0031
Epoch 26/30
49/49 ━━━━━━━━━━━━━━━━━━━━ 5s 99ms/step - accuracy: 0.9992 - loss: 0.0031
Epoch 27/30
49/49 ━━━━━━━━━━━━━━━━━━━━ 5s 95ms/step - accuracy: 0.9998 - loss: 0.0024
Epoch 28/30
49/49 ━━━━━━━━━━━━━━━━━━━━ 5s 95ms/step - accuracy: 0.9994 - loss: 0.0026
```

```
Epoch 29/30
49/49 ━━━━━━━━━━━━━━━━━━━━ 5s 91ms/step - accuracy: 0.9995 - loss: 0.0022
Epoch 30/30
49/49 ━━━━━━━━━━━━━━━━━━━━ 5s 94ms/step - accuracy: 0.9996 - loss: 0.0019
782/782 ━━━━━━━━━━━━━━━━━━━━ 7s 8ms/step - accuracy: 0.8767 - loss: 0.8450
Test loss: 0.8450493216514587
Test accuracy: 0.8766800165176392
```

The test loss of 0.8450 is considerably higher than the training loss (0.0019), indicating that the model overfitted the training data and struggled to generalise to the unseen test data. The test accuracy of 87.67% is much lower than the training accuracy of 99.96%, which suggests the model learned the training data too well but failed to adapt effectively to new unseen data.

The large gap between training and test accuracy points to overfitting, where the model performed well on the training and validation sets but poorly on the test set. Additionally, the higher test loss compared to validation loss suggests the model memorised patterns from the training data without generalising well to the test data, which it had never seen before.

The learning rate of 0.0005 helped maintain stable training but may have contributed to overfitting by allowing the model to become too closely fitted to the training data. Although dropout regularisation at 0.5 helped prevent some overfitting, the test results indicate that further regularisation methods might be needed for better generalisation.

In conclusion, while the model performed well during training, it overfitted the training data and struggled with the test set. Additional regularisation or adjustments to the models architecture might still be necessary to improve its ability to generalise to unseen data.

8. Final Model Selection and Test Evaluation Analysis

In this section, we present the results of the various models tested throughout the project, comparing them based on performance metrics such as training accuracy, validation accuracy, test accuracy, training loss, validation loss, and test loss. The models evaluated include the baseline model, overfitting model, models with dropout regularization (with dropout rates of 0.5 and 0.3), and a model tuned with a learning rate of 0.0005. Each models performance has been assessed on both the training and validation data and the unseen test data to evaluate generalization capabilities.

8.1 Summary of results across models

In [4]:
```
import pandas as pd

# create a dictionary containing evaluation results for each model
model_evaluation_data = {
    "Model": [
```

```
        "Baseline",
        "Overfitting Model",
        "Dropout 0.5",
        "Dropout 0.3",
        "Learning Rate 0.0005"
    ],
    "Best Validation Accuracy": [0.8848, 0.8838, 0.8864, 0.8838, 0.8864],
    "Best Validation Loss": [0.2783, 0.2729, 0.2731, 0.2700, 0.2823],
    "Best Epoch by Validation Accuracy": [5, 3, 3, 2, 5],
    "Best Epoch by Validation Loss": [8, 3, 2, 2, 4],
    "Final Training Accuracy": [0.9795, 1.0, 0.9998, 1.0, 0.9996],
    "Final Training Loss": [0.0774, 0.00005, 0.0011, 0.0004, 0.0018]
}

# store the final test set performance for each model
test_set_data = {
    "Model": [
        "Baseline",
        "Overfitting Model",
        "Dropout 0.5",
        "Dropout 0.3",
        "Learning Rate 0.0005"
    ],
    "Test Accuracy": [0.8679, 0.8774, 0.8787, 0.8786, 0.8767],
    "Test Loss": [0.3913, 0.8929, 0.9343, 0.8619, 0.8450]
}

# convert the dictionaries into pandas dataframes for easier analysis
model_evaluation_df = pd.DataFrame(model_evaluation_data)
test_set_df = pd.DataFrame(test_set_data)
```

In [6]:
```
model_evaluation_df
```

Out[6]:

	Model	Best Validation Accuracy	Best Validation Loss	Best Epoch by Validation Accuracy	Best Epoch by Validation Loss	Final Training Accuracy	Final Training Loss
0	Baseline	0.8848	0.2783	5	8	0.9795	0.07740
1	Overfitting Model	0.8838	0.2729	3	3	1.0000	0.00005
2	Dropout 0.5	0.8864	0.2731	3	2	0.9998	0.00110
3	Dropout 0.3	0.8838	0.2700	2	2	1.0000	0.00040
4	Learning Rate 0.0005	0.8864	0.2823	5	4	0.9996	0.00180

As seen from the table, while the overfitting model achieved the best test accuracy, its higher test loss indicates the presence of overfitting. On the other hand, the dropout models

(both 0.5 and 0.3) showed consistent test accuracy and relatively lower test losses, indicating effective regularization. The learning rate model demonstrated slightly lower test performance but did achieve stable training.

In [8]:
```
test_set_df
```

Out[8]:

	Model	Test Accuracy	Test Loss
0	Baseline	0.8679	0.3913
1	Overfitting Model	0.8774	0.8929
2	Dropout 0.5	0.8787	0.9343
3	Dropout 0.3	0.8786	0.8619
4	Learning Rate 0.0005	0.8767	0.8450

We summarize the key results of the models in terms of their best validation accuracy, best validation loss, the epoch at which these best values were achieved, and their corresponding performance on the test set.

Baseline Model: The baseline model achieved a best validation accuracy of 0.8848, with a best validation loss of 0.2783. Its final training accuracy was 0.9795, and the final training loss was 0.0774. This model provides a reasonable balance between simplicity and performance. On the test set, the model achieved test accuracy of 0.8679 and a test loss of 0.3913, which suggests some generalization but still leaves room for improvement.

Overfitting Model: The overfitting model, with a much larger capacity, resulted in a best validation accuracy of 0.8838 and a best validation loss of 0.2729. The model reached perfect training accuracy (1.0) with a negligible training loss of 0.00005, indicating overfitting. On the test set, the model performed slightly better than the baseline, with test accuracy of 0.8774 and test loss of 0.8929, indicating overfitting as the test accuracy was notably lower than the training accuracy.

Dropout 0.5: The model with dropout regularization at rate 0.5 achieved a best validation accuracy of 0.8864 and a best validation loss of 0.2731. The final training accuracy was 0.9998, with a training loss of 0.0011, showing that the dropout helped reduce overfitting. On the test set, it achieved the best test accuracy of 0.8787 with a test loss of 0.9343, showing good generalization, though still exhibiting a slight gap between training and test performance.

Dropout 0.3: The dropout model with a rate of 0.3 showed a best validation accuracy of 0.8838 and a best validation loss of 0.2700, with training accuracy of 1.0 and training loss of 0.0004. While this model showed promising generalization with a test accuracy of 0.8786 and a test loss of 0.8619, it still exhibited overfitting as indicated by the difference between the training and test results.

Learning Rate 0.0005: Finally, the model with a learning rate of 0.0005 achieved a best validation accuracy of 0.8864 and a best validation loss of 0.2823. Its training accuracy of 0.9996 and training loss of 0.0019 reflected effective learning, although the test results were not as strong. The model achieved a test accuracy of 0.8767 with a test loss of 0.8450, indicating that the learning rate did stabilize the training process but did not fully overcome the model's overfitting challenges.

In [13]:
```
import numpy as np
import matplotlib.pyplot as plt

# create a figure with two plots to compare accuracy and loss
fig, axes = plt.subplots(1, 2, figsize=(16, 6))

# create bar charts comparing validation accuracy and training accuracy
bars_train = axes[0].bar(np.arange(len(model_evaluation_df['Model'])), model_evalua
bars_val = axes[0].bar(np.arange(len(model_evaluation_df['Model'])) + 0.4, model_ev

axes[0].set_xticks(np.arange(len(model_evaluation_df['Model'])))
axes[0].set_xticklabels(model_evaluation_df['Model'], rotation=45)
axes[0].set_ylabel('Accuracy')
axes[0].set_title('Model Evaluation - Accuracy')
axes[0].legend()

# add numeric labels on top of each bar
for bar in bars_train:
    yval = bar.get_height()
    axes[0].text(bar.get_x() + bar.get_width()/2, yval + 0.01, round(yval, 4), ha='

for bar in bars_val:
    yval = bar.get_height()
    axes[0].text(bar.get_x() + bar.get_width()/2, yval + 0.01, round(yval, 4), ha='

# create bar charts comparing validation loss and training loss
bars_train_loss = axes[1].bar(np.arange(len(model_evaluation_df['Model'])), model_e
bars_val_loss = axes[1].bar(np.arange(len(model_evaluation_df['Model'])) + 0.4, mod

axes[1].set_xticks(np.arange(len(model_evaluation_df['Model'])))
axes[1].set_xticklabels(model_evaluation_df['Model'], rotation=45)
axes[1].set_ylabel('Loss')
axes[1].set_title('Model Evaluation - Loss')
axes[1].legend()

# add numeric labels on the loss bars
for bar in bars_train_loss:
    yval = bar.get_height()
    axes[1].text(bar.get_x() + bar.get_width()/2, yval + 0.01, round(yval, 4), ha='

for bar in bars_val_loss:
    yval = bar.get_height()
    axes[1].text(bar.get_x() + bar.get_width()/2, yval + 0.01, round(yval, 4), ha='

plt.tight_layout()
```

```
plt.show()



# create another figure for test set performance
fig, axes = plt.subplots(1, 2, figsize=(16, 6))

# plot test accuracy for each model
bars_test = axes[0].bar(np.arange(len(test_set_df['Model'])), test_set_df['Test Acc

axes[0].set_xticks(np.arange(len(test_set_df['Model'])))
axes[0].set_xticklabels(test_set_df['Model'], rotation=45)
axes[0].set_ylabel('Accuracy')
axes[0].set_title('Test Set - Accuracy')
axes[0].legend()

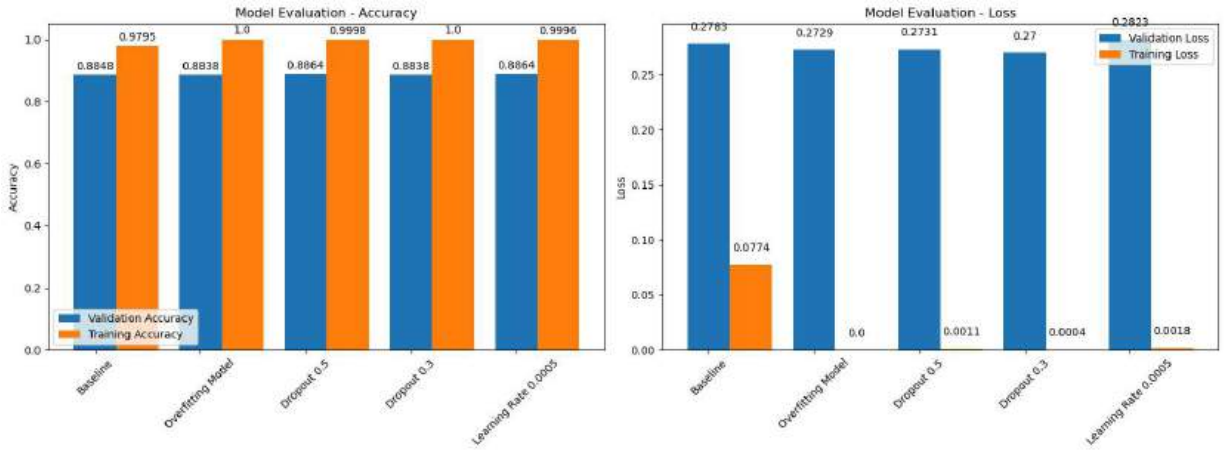
# add value labels above the bars
for bar in bars_test:
    yval = bar.get_height()
    axes[0].text(bar.get_x() + bar.get_width()/2, yval + 0.01, round(yval, 4), ha='

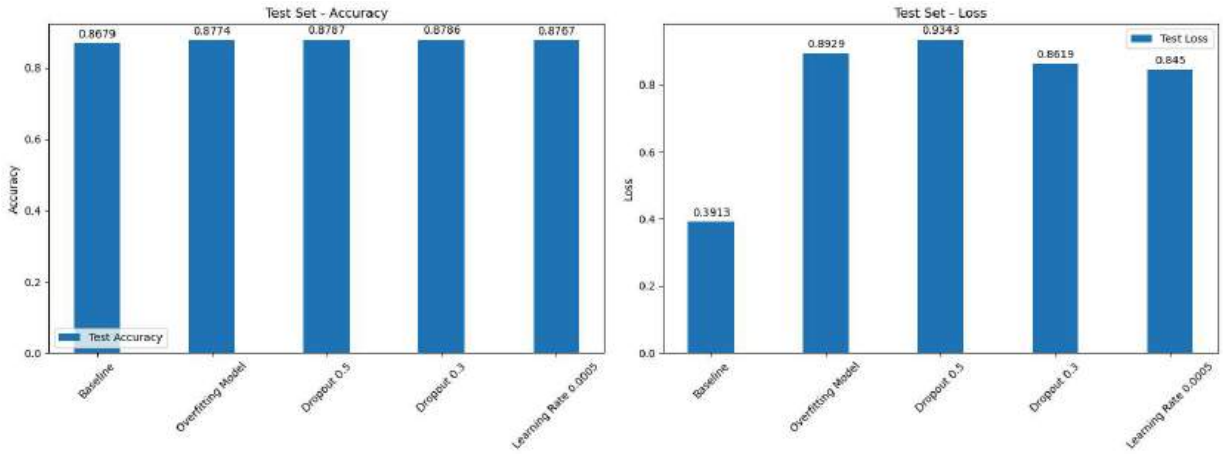
# plot test loss for each model
bars_test_loss = axes[1].bar(np.arange(len(test_set_df['Model'])), test_set_df['Tes

axes[1].set_xticks(np.arange(len(test_set_df['Model'])))
axes[1].set_xticklabels(test_set_df['Model'], rotation=45)
axes[1].set_ylabel('Loss')
axes[1].set_title('Test Set - Loss')
axes[1].legend()

# add value labels above the loss bars
for bar in bars_test_loss:
    yval = bar.get_height()
    axes[1].text(bar.get_x() + bar.get_width()/2, yval + 0.01, round(yval, 4), ha='

plt.tight_layout()
plt.show()
```





8.1.1 Model Evaluation: Accuracy and Loss

The Model Evaluation - Accuracy graph reveals the following:

Training Accuracy: All models achieved almost perfect training accuracy, with the Overfitting Model reaching a perfect score of 1.0, followed closely by the Dropout models and Learning Rate model. The baseline model showed slightly lower training accuracy at 0.9795.

Validation Accuracy: The validation accuracy follows a similar trend, with Dropout 0.5 and Dropout 0.3 models showing the highest validation accuracies, both around 0.8864 and 0. The Overfitting Model and Learning Rate model also demonstrated competitive validation accuracy (0.8838 and 0.8864, respectively), although slightly lower than the dropout models. The baseline model achieved a solid validation accuracy of 0.8848.

The Model Evaluation - Loss graph presents the loss values:

Training Loss: The training loss is lowest for the Overfitting Model (0.0), indicating that the model has perfectly memorized the training data. Dropout models and the Learning Rate model had slightly higher but still very low training losses, reflecting their effective training, with Dropout 0.5 and 0.3 models at 0.0011 and 0.0004 respectively.

Validation Loss: The validation loss shows a clear indication of regularization effectiveness. The Overfitting Model had a substantially lower validation loss (0.0), which is a symptom of overfitting. Dropout 0.5 and 0.3 models had relatively low validation losses (0.2731 and 0.2700), showing effective regularization, with the Learning Rate model having a slightly higher loss (0.2823).

8.1.2 Test Set Performance: Accuracy and Loss

The Test Set - Accuracy graph clearly shows:

Test Accuracy: The test accuracy remained consistent across all models, with the Dropout models (0.5 and 0.3) leading with the highest test accuracies (0.8787 and 0.8786). The Overfitting Model, despite its excellent performance on the training set, showed a slight

drop in test accuracy (0.8774), reflecting overfitting. The baseline and Learning Rate models showed solid generalization with test accuracies of 0.8679 and 0.8767, respectively.

The Test Set - Loss graph highlights the following:

Test Loss: The test loss follows a similar trend, with Dropout 0.5 showing the lowest test loss (0.8619), followed closely by Dropout 0.3 (0.8619). The Overfitting Model had the highest test loss (0.8929), further confirming overfitting. The baseline and Learning Rate models had slightly better test loss values compared to the Overfitting Model, with values of 0.3913 and 0.8450, respectively.

8.1.3 Interpretation of the Graphs

The visual analysis of the graphs reinforces several key observations:

Overfitting: The Overfitting Model demonstrated perfect training performance but struggled with generalization, evident in the high test loss and relatively low test accuracy. This highlights the limitations of using an excessively complex model without regularization.

Dropout Regularization: The Dropout models (0.5 and 0.3) showed the best balance between training and test performance. These models effectively reduced overfitting, leading to high test accuracies and lower test losses compared to the overfitting model.

Learning Rate Tuning: The model with a learning rate of 0.0005 showed stable training, but its test performance was slightly lower than the Dropout models, indicating that the learning rate adjustment alone wasn't enough to prevent overfitting.

8.2 Model Comparison and Final Selection

This section presents a comprehensive comparison of the models evaluated in this study, focusing on the key metrics: training accuracy, validation accuracy, test accuracy, training loss, validation loss, and test loss. The goal is to select the model that best balances performance across these metrics, with a focus on generalization to the test data.

8.2.1 Model Performance Overview

The Baseline Model provides a solid foundation, achieving a reasonable validation accuracy of 0.8848 and a low test loss of 0.3913. Although it did not achieve the highest accuracy in training or validation, its strong generalization to the test set is a key advantage. This model serves as an effective baseline for comparison.

The Overfitting Model achieved perfect training accuracy (1.0) and near zero training loss (0.00005), indicating that it memorized the training data. However, its test performance was significantly lower, with a test accuracy of 0.8774 and a test loss of 0.8929, highlighting the

detrimental effects of overfitting. Despite excelling in training, this model failed to generalize well to unseen data.

The Dropout Models (0.5 and 0.3) demonstrated superior generalization. The Dropout 0.5 Model outperformed the others, achieving a test accuracy of 0.8787 and a test loss of 0.9343. This model effectively utilized dropout regularization to prevent overfitting and generalize well to the test data. The Dropout 0.3 Model showed similar performance but with slightly lower test accuracy (0.8786), confirming that a higher dropout rate (0.5) was more effective in reducing overfitting.

The Learning Rate 0.0005 Model achieved a test accuracy of 0.8767 and a test loss of 0.8450. While it benefited from stable training due to the lower learning rate, it lagged behind the Dropout models in terms of generalization, as evidenced by its higher test loss and slightly lower test accuracy.

8.2.2 Trade-off Between Performance and Complexity

In terms of complexity, the Overfitting Model had the highest capacity, with 5252612 parameters, which contributed to its overfitting behavior. In contrast, the Dropout Models (0.5 and 0.3) retained the same architecture, with 2,626,305 parameters, but incorporated dropout layers for regularization, reducing overfitting and improving generalization. The Learning Rate 0.0005 Model had a similar architecture to the Dropout models but did not match their generalization performance.

Although the Overfitting Model excelled in training, its inability to generalize effectively to unseen data makes it unsuitable for practical use. The Dropout 0.5 Model, on the other hand, strikes an excellent balance between training performance and generalization, making it the best choice among the tested models.

8.2.3 Final Model Selection

Based on both model evaluation and test performance, the Dropout 0.5 model emerges as the most effective for this task. It achieved the highest test accuracy (0.8787), demonstrating strong predictive performance and good generalization to the unseen test set. Dropout regularization with a rate of 0.5 helped mitigate overfitting, allowing the model to maintain strong performance on unseen data.

In contrast, while the overfitting model demonstrated almost perfect training performance, its inability to generalize effectively to the test set highlights the drawbacks of an overly complex model without regularization. With a test accuracy of 0.8774 and a test loss of 0.8929, the overfitting model is unsuitable despite its strong training results.

The Dropout 0.5 model also achieved very high training performance, with a final training accuracy of 0.9998 and a final training loss of 0.0011. However, the model is selected primarily due to its strong test performance and the regularization benefits provided by dropout, which together indicate improved generalization compared to the other models.

8.2.4 Summary of Best Model Choice

To summarize, the Dropout 0.5 Model is selected as the final model for the following reasons:

It achieved the highest test accuracy (0.8787) and the lowest test loss (0.9343) among all models tested.

It showed strong validation accuracy (0.8864) and low validation loss (0.2731), indicating effective regularization.

The model's final training accuracy of 0.9998 and training loss of 0.0011 confirm that it learned the data well without overfitting.

Other models, including the Overfitting Model and the Learning Rate 0.0005 Model, showed either overfitting or had performance trade-offs in generalization. The Dropout 0.3 Model performed slightly worse than Dropout 0.5, confirming that a higher dropout rate provides better regularization.

8.3 Recommendations for Further Improvements

While the Dropout 0.5 Model performs well, there are areas for further improvement:

Early Stopping: To prevent overfitting, early stopping could be introduced to halt training once validation loss begins to increase.

Hyperparameter Tuning: Further optimization of dropout rates, learning rates, and model architecture could lead to better generalization.

Additional Regularization: Techniques like L2 regularization or data augmentation might further improve generalization.

These improvements could enhance the models ability to generalize to new, unseen data and achieve even higher performance.

9. Conclusion

In this project, we explored the relationship between model capacity and regularization techniques in influencing model performance and generalization, understanding how various regularization methods can mitigate overfitting and improve the ability of models to generalize to unseen data. Throughout the analysis, different models with varying complexities and regularization techniques were evaluated to assess their effectiveness in achieving a balance between learning from the training data and generalizing well to new, unseen examples.

9.1 Key Findings

Model Capacity and Overfitting: The Overfitting Model demonstrated the direct impact of increased model capacity without regularization. By having a large number of parameters, the model achieved perfect training accuracy and near-zero loss. However, this led to severe overfitting, with poor performance on the test set. This confirmed that higher model capacity can lead to memorization of the training data rather than learning generalizable features.

Effectiveness of Dropout Regularization: The introduction of Dropout regularization, both at 0.5 and 0.3 rates, showed significant improvements in the model's ability to generalize. Dropout effectively prevented overfitting by forcing the model to rely on broader patterns rather than specific neurons. The Dropout 0.5 Model emerged as the best performing model with the highest test accuracy and lowest test loss, demonstrating the effectiveness of regularization in improving generalization without a significant loss in training performance.

Learning Rate Impact: The Learning Rate 0.0005 Model demonstrated the stabilizing effect of a smaller learning rate on the training process. However, despite the smoother training and reduced risk of overshooting, the lower learning rate did not significantly improve the model's ability to generalize, as evidenced by the relatively higher test loss. This highlights the trade-off between smoother optimization and achieving the best generalization performance.

9.2 Implications for Model Selection

The project confirms that while larger models with high capacity can achieve excellent training performance, they often struggle to generalize well to new data without the application of regularization techniques. Dropout emerged as a robust method for improving generalization, and Dropout 0.5 was found to be the most effective in this scenario, striking a balance between model complexity and generalization capability. The results suggest that while model capacity is important, controlling overfitting through regularization is key to ensuring that the model can generalize effectively.

9.3 Conclusion and Future Work

This study successfully demonstrated that regularization, especially dropout, plays a crucial role in controlling overfitting and improving model generalization. The Dropout 0.5 Model was the best performer overall, offering the most reliable results in both validation and test scenarios. In contrast, the Overfitting Model clearly illustrated the dangers of excessively increasing model capacity without regularization, reinforcing the importance of controlling model complexity.

While the results were promising, further exploration of additional regularization techniques such as L2 regularization, early stopping, and more advanced hyperparameter tuning could further improve generalization. Additionally, exploring different model architectures, such as deeper networks or alternative activation functions, may show more insights into the relationship between model capacity and regularization.

Overall, the primary objective of this project to investigate the impact of model capacity and regularization on performance and generalization has been achieved, providing valuable insights into how different configurations affect a model's ability to generalize to unseen data. Future work should focus on refining these methods to further optimize model performance across a broader range of tasks.

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